The one-copy Bernoulli convolution at 2/3: an L^2 dictionary, two theorems, and the routes it closes

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Abstract

Let $Y = \frac{1}{3}\sum_{k\ge 0} B_k (2/3)^k$ with B_k fair independent bits, and let μ be its law. Whether μ is absolutely continuous is a long-standing open question. We record an L^2 route: an elementary reduction, two theorems that make the independence heuristics of the subject exact, an unconditional partial-sum bound, and proofs that several natural ways of finishing cannot work. Nothing here settles the question, and we say precisely where each attempt stops. One route is not closed. Recast as a charge on the runs of the exact digit recursion $2A_{n+1} - 3A_n \in \{0, \pm 1, \pm 2\}$, a strictly local and finite-order combinatorial cost yields a bound on the octave energy that decays, at $R^{-0.28}$ over nineteen octaves by exact enumeration, where every previously tested cost grows; making that a theorem requires a large-deviation estimate for the run statistics of the $\times 3/2$ orbit, and we say what is measured and what is not. What survives is a single statement about $I_K = \int_1^2 \prod_{m\le K}\bigl(1 + \cos(2\pi x (3/2)^m/3)\bigr)dx$, the integral of a Riesz product with the non-Pisot ratio $3/2$ over a single octave: membership in L^2 is equivalent to $\sum_K (3/4)^K I_K < \infty$, so what has to be excluded is growth at rate $4/3$ per step rather than mere unboundedness. Boundedness of I_K is sufficient and strictly stronger. The lattice, the window position and every Diophantine property of $(3/2)^K$ disappear from the statement; that equivalence is measured here rather than proved, and we say what its proof would require. Throughout, claims are tested by substituting a biased coin, which leaves the ratio, the lattice and the period unchanged but makes the measure singular for a small enough bias: an argument that survives that substitution is proving too much. The measure itself, its characteristic function as an infinite product, its self-similarity and the functional equation are formalised in Lean 4 with Mathlib, as are the two theorems and the arithmetic lemma behind them.

1 The reduction to L^2

Write μ_n for the law of $\{2^n Y\}$. For $U \sim \mu_n$ put $2U = H + Z$ with $H = \lfloor 2U \rfloor$, $u_n(Z) = \mathbb{E}[(-1)^H \mid Z]$ and $j_n = \log 2 - H(H \mid Z)$. The j_n are the fibre deficits, and $\sum_n j_n$ is the relative entropy of μ against Lebesgue measure, so summability is equivalent to μ having a density of finite entropy.

Two remarks. First, μ is symmetric about $1/2$, so $\mu_n([0, \frac{1}{2})) = \frac{1}{2}$ at every level and $j_n = I(H_n; Z_n)$ is a mutual information: the question is whether the total information the binary digits of Y carry about the tail is finite. Second, with f the density and $A = \int f^2$, the inequality $\log t \le t - 1$ gives $\int f \log f - \log A = \int f \log(f/A) \le \int f(f/A - 1) = 0$, so

$$\sum_n j_n = \int f \log f \;\le\; \log\Bigl(1 + 2\sum_{k\ge 1} |\hat{\mu}(k)|^2\Bigr). \tag{1}$$

An L^2 density therefore gives summable deficits outright. The target becomes $S = \sum_{k\geq 1} |\hat{\mu}(k)|^2 < \infty$, and numerically $S = 0.103155$, so that $\|f\|_2^2 = 1 + 2S = 1.206310$.

2 Two exact identities

Since $3Y = B_0 + 2Y'$ with Y' an independent copy, $\{3Y\}$ and $\{2Y\}$ have the same law, so

$$\hat{\mu}(3k) = \hat{\mu}(2k) \qquad \text{for every } k \in \mathbb{Z}. \tag{2}$$

Writing $k = 2^a 3^b m$ with $(m, 6) = 1$ and applying (2) b times gives $\hat{\mu}(2^a 3^b m) = \hat{\mu}(2^{a+b} m)$: the transform is constant on the anti-diagonals of the 2–3 lattice, whence

$$\sum_{k\geq 1} |\hat{\mu}(k)|^2 = \sum_{(m,6)=1} \sum_{s\geq 0} (s+1)\, |\hat{\mu}(2^s m)|^2. \tag{3}$$

Membership in L^2 is thus a lacunary sum with a linear multiplicity weight. Numerically (2) holds to $9 \cdot 10^{-16}$ for $k \leq 4000$. One practical consequence: within an octave the dominant frequencies occur as lattice twins with identical values, for instance $108 = 2^2 3^3$ and $72 = 2^3 3^2$, so neighbouring octaves are not independent.

In continuous form $\hat{\mu}(\xi) = c(\xi)\hat{\mu}(2\xi/3)$ with $c(\xi) = (1+e^{-2\pi i\xi/3})/2$ and $|c(\xi)|^2 = \cos^2(\pi\xi/3)$. Substituting $\eta = 2\xi/3$ in $\int_{\mathbb{R}} |\hat{\mu}|^2$ and applying Plancherel gives, whenever the density lies in L^2,

$$\|f\|_2^2 = 3 \int f(x) f(x + \tfrac{1}{2})\, dx. \tag{4}$$

Measured, 1.206310 against 3×0.402103, agreeing to $6 \cdot 10^{-7}$; the same relation applied to the folded measure gives 1.005 against 1.496 and fails, so (4) is not an artefact. Iterating the functional equation produces a family of such exact relations among finitely many autocorrelation values, the next being $\frac{7}{9}A(0) = A(\frac{1}{2}) + A(\frac{3}{4}) + \frac{1}{2}A(\frac{1}{4})$.

Summing (2) over k gives one more, on the spectral side. The multiples of 3 carry $\sum_{j\geq 1} |\hat{\mu}(3j)|^2 = \sum_{j\geq 1} |\hat{\mu}(2j)|^2$, which is the energy on the even indices, so subtracting from S in two ways,

$$\sum_{3\nmid k} |\hat{\mu}(k)|^2 = \sum_{k \text{ odd}} |\hat{\mu}(k)|^2 :$$

the energy on indices coprime to 3 equals the energy on odd indices. Both sides measure 0.100526 to a relative $9 \cdot 10^{-8}$ at $N = 3 \cdot 10^6$, and the functional equation itself holds termwise to $1.7 \cdot 10^{-15}$ over the same range. The rearrangement with the $(s+1)$ weight reproduces $S = 0.103155$ over exactly the $2 \cdot 10^6$ indices coprime to 3 below N, so the multiplicity is counted once and not twice, which is the obvious way for this bookkeeping to be wrong. Finally the two sides of the dictionary meet: $1 + 2S = 1.206310$ against the meet-in-the-middle $\|f\|_2^2 = 1.206310$, computed with no shared step.

3 The seed form and two theorems

For $\xi \geq 3$ there is a unique K with $x = \xi(2/3)^K/3 \in [1, 3/2)$, and the product splits there:

$$|\hat{\mu}(\xi)| = P_K(x)\, T(x) \leq P_K(x), \qquad P_K(x) = \prod_{j=0}^{K} \bigl|\cos(\pi(3/2)^j x)\bigr|, \quad T(x) = \prod_{m\geq 1} \bigl|\cos(\pi x(2/3)^m)\bigr|. \tag{5}$$

The tail T does not tend to 1: on $[1, 3/2]$ it has supremum 0.046, mean square $1.11 \cdot 10^{-3}$, and a double zero at $x = 9/8$. Only the inequality is used below, which is the safe direction; a constant read off as if (5) were an equality is wrong by three orders of magnitude, though the exponents are unaffected. Equivalently, for integers k, $|\hat{\mu}(k)|^2 = \prod_{i \geq 0} \cos^2(\pi k 2^i / 3^{i+1})$, each factor depending only on $k \bmod 3^{i+1}$.

Lemma 1. *Let $n_i = 2^i 3^{K-i}$ for $0 \leq i \leq K$. If $\varepsilon_i \in \{0, \pm 1, \pm 2\}$ and $\sum_i \varepsilon_i n_i = 0$, then $\varepsilon_i = 0$ for every i.*

Proof. Otherwise let i_1 be the largest index with $\varepsilon_{i_1} \neq 0$. As ε_{i_1} is prime to 3, the 3-adic valuation of $\varepsilon_{i_1} n_{i_1}$ is exactly $K - i_1$, while every other nonzero term has index $i < i_1$ and valuation $K - i > K - i_1$. A sum whose terms have a unique minimal valuation has that valuation, so it is nonzero. □

The frequencies are therefore dissociate; an exhaustive search finds no $\{0, \pm 1, \pm 2\}$ relation for $K \leq 8$.

Theorem 2. *For every K,*

$$3^{-(K+1)} \sum_{k \bmod 3^{K+1}} \prod_{i=0}^{K} \cos^2\Big(\frac{\pi k 2^i}{3^{i+1}}\Big) = 2^{-(K+1)}, \tag{6}$$

exactly.

Proof. Expand $\cos^2(\pi\theta) = (1 + \cos 2\pi\theta)/2$. Averaging over a full period retains exactly those $\varepsilon \in \{0, \pm 1\}^{K+1}$ with $3^{K+1} \mid \sum_i \varepsilon_i n_i$; by the valuation count of Lemma 1 that sum has valuation $K - i_1 \leq K < K + 1$ whenever $\varepsilon \neq 0$. Only $\varepsilon = 0$ survives, contributing $2^{-(K+1)}$. □

Verified to ten decimals for $K = 0, \ldots, 9$. The independence heuristic behind estimates of this kind is thus exact at the level of full periods, with no error term.

Remark. Lemma 1, the orthogonality relation it is combined with, the expansion of the product, and their assembly are formalised in Lean 4 with Mathlib in `BC23Full.lean`, which compiles with no `sorry`; every theorem there prints only `propext`, `Classical.choice` and `Quot.sound`. Three further files carry the rest of the chain, also `sorry`-free with the same axioms: `BC23Entropy.lean` for the reduction (1), in both its integral and its finite-sum form; `BC23PartialSum.lean` for Proposition 3, derived from the period bound taken as a hypothesis, with the constant 3/2 shown sharp by a worst-case aggregation check; and `BC23StepContraction.lean` for the step and two-level contraction implications.

A further file has the measure itself, which the others do not: `BC23Measure.lean` builds μ as the pushforward of the infinite product of fair Bernoulli measures under Y, and derives $\mu([0,1]) = 1$, the characteristic function as a limit of finite products, the product formula $|\hat{\mu}(\xi)| = \lim_n \big|\prod_{k<n} \cos(\pi\xi(2/3)^k/3)\big|$, the self-similarity $\hat{\mu}(\xi) = \frac{1}{2}\big(1 + e(-\xi/3)\big)\hat{\mu}(2\xi/3)$, and (2) for integer arguments. Everything before this was stated about an assumed density; these are statements about μ.

The eight files were recompiled as a set on 8 September against the same Mathlib: 39 certified statements, no `sorry`, no `sorryAx` in any axiom report, no errors, and only `propext`, `Classical.choice` and `Quot.sound` throughout. As a control that the check can fail, a copy of `BC23Measure.lean` with (2) altered to $\hat{\mu}(3m) = \hat{\mu}(5m)$ does not typecheck.

Proposition 3. *For every $M \geq 1$, $\sum_{k=1}^{M} |\hat{\mu}(k)|^2 \leq \frac{3}{2} M^{\log(3/2)/\log 3} = \frac{3}{2} M^{0.36907}$.*

Proof. Every factor is at most one, so $|\hat{\mu}(k)|^2 \leq F_I(k) := \prod_{i=0}^{I} \cos^2(\pi k 2^i / 3^{i+1})$ for any I. By Theorem 2 the sum of F_I over a full period 3^{I+1} is $(3/2)^{I+1}$. Take $I + 1 = \lceil \log_3 M \rceil$ and use nonnegativity. □

In correlation-dimension terms this reads $D_2 \geq 1 - \log(3/2)/\log 3 = 0.63093$. It is a power saving over the $M(\log M)^{-0.46}$ that the proved logarithmic Fourier decay at a rational non-Pisot parameter would give, and it is comprehensively superseded, which should be said plainly. A two-line folklore argument already gives $D_2 \geq \log 2/\log(9/4) = 0.85476$; Kleptsyn, Pollicott and Vytnova prove rigorously, by a computer-assisted method stated for exactly representable rational parameters, that $D_2(\mu_\lambda) \geq 0.97580$ for every $\lambda \in [0.617, 0.743]$, an interval containing $2/3$, with a refined pass giving 0.9900 on $[0.5703, 0.8)$; and Shmerkin's L^q-dimension theorem applies verbatim here, with exponential separation $\Gamma_k = 3^{-(k+1)}$ and an exceptional set containing no rational, giving $D_2 = 1$ outright, that is $\sum_{|k| \leq R} |\hat{\mu}(k)|^2 = O_\varepsilon(R^\varepsilon)$, though with ineffective constants. Proposition 3 is therefore a short self-contained derivation and nothing more; it is recorded because it follows from Theorem 2 in four lines, not because it is competitive. The three comparisons in this paragraph come from a literature search and I have not read the primary sources myself; the numbers should be checked against them before being relied on.

4 What remains, and what is closed

The octave statement is not new terminology. That $\sum_{R \leq k < 2R} |\hat{\mu}(k)|^2 \leq C R^{-\delta}$ for all R is, up to ε, Sobolev regularity: $\mu \in L^2_\sigma$ for every $\sigma < \delta/2$, equivalently Sobolev dimension exceeding 1 by δ. For a rational non-Pisot parameter nothing of the kind is published, not even $\delta = 0$, and absolute continuity is open for every rational $p/q \in (1/2, 1)$ with $p \geq 2$.

It is worth recording exactly why the modern criteria do not reach $2/3$, because the answer is sharper than "not yet". Kittle's theorem (arXiv:2103.12684, Ann. Sci. ENS 57 (2024) 1191–1230) asks for $(d \log M_F - h_F)(\log M_F)^2 < \frac{1}{27}(\log M_F - \log \lambda^{-1})^3 \lambda^2$. At $2/3$ every input is exact: the entropy is maximal, $h_F = \log 2$, since the digit map is injective; $M_F = M_\lambda = 3$, the minimal nonzero $|\nu|$ being 1; and $\log M_F - h_F = \log(3/2)$ with $\log M_F - \log \lambda^{-1} = \log 2$. The left side is 0.4893757 and the right 0.0054819, so the criterion fails by a factor 89.27. That is not a near miss. The condition is equivalent to $M_\lambda < F(\lambda)$ with F increasing and bounded by $2^{27/26} = 2.0540$, while every rational p/q in $(1/2, 1)$ has $M = q \geq 3$: Kittle's criterion can reach no rational parameter at all. Varju's theorem (arXiv:1602.00261, JAMS 32 (2019)) does admit rationals, but only for $1 - \lambda < 7 \cdot 10^{-37}$, against $1 - \lambda = \frac{1}{3}$ here.

Kittle and Kogler (arXiv:2409.18936, Cor. 1.11) leave the question in an unusually concrete state. Their condition $(1 - \lambda)^{-1} h_\lambda > C\big(\max\{1, \log\log M/h\}\big)^2$ collapses at $2/3$, where the maximum is 1, to the single requirement $C < 3 \log 2 = 2.0794$ on one absolute constant; their own unoptimised estimate is $C \approx 10^{10}$. In the other direction the golden ratio, where the measure is singular, forces $C \geq 1.2544$. So absolute continuity at $2/3$ sits inside a factor 1.66 of one constant in an existing framework, with about nine of those ten orders of magnitude being machinery loss rather than mathematics. Provenance, since it matters here: these papers were read and every number re-derived by two independent passes, but not by me, and I have not read the primary sources myself; the 10^{10} is the authors' own estimate of their own method

and is not a theorem, resting as it does on an unspecified absolute constant from Breuillard–Varjú. The claim that no published criterion reaches $2/3$ is a literature-wide negative built on four papers read and three abstracts, and should be treated as the current state of a search rather than as a fact.

An octave $[R, 2R)$ supplies about $R = (3/2)^K$ consecutive integers, a fraction of order $R^{-1.71}$ of the period 3^{K+1}. Through the character expansion,

$$\sum_{k\in W} F_K(k) = 2^{-(K+1)}\big(L + \mathrm{Err}\big), \qquad \mathrm{Err} = \sum_{\varepsilon\neq 0} c_\varepsilon D(\nu_\varepsilon), \tag{7}$$

with $c_\varepsilon = 2^{-|\operatorname{supp}\varepsilon|}$, $\nu_\varepsilon = \sum_i \varepsilon_i n_i$, $|D(\nu)| \le \min(L, P/2|\nu|)$. The main term alone gives $E(R) \ll R^{-0.71}$, which suffices, since S is a sum over octaves and $E(R) \le CR^{-\delta}$ for any $\delta > 0$ is enough. Everything turns on Err.

The image, and how the weight sits on it. The map $\varepsilon \mapsto \nu_\varepsilon$ is injective on $\{-1, 0, 1\}^{K+1}$, and its image contains every integer with $|m| \le 2^{K+1} - 1$, the full range being $|\nu_\varepsilon| \le \sum_i n_i = 3^{K+1} - 2^{K+1}$. Uniqueness is the same induction as Lemma 1 read as a digit algorithm: modulo 3 only the $i = K$ term survives and $\{-1, 0, 1\}$ is a complete residue system, so the top digit is forced; divide by 3 and repeat. Verified exhaustively for $K \le 9$. Writing $c(m) = 2^{-\#\{\text{nonzero digits of } m\}}$ for the forced expansion, (7) becomes $\mathrm{Err} = 2\sum_{m\ge 1} c(m) \operatorname{Re} D_W(m)$, an arithmetically weighted sum with $\sum_m c(m) = 2^{K+1} - 1$. That total runs over the whole image, which is spread over $|m| \le 3^{K+1} - 2^{K+1}$ and is not consecutive; the consecutive block $|m| \le 2^{K+1} - 1$ is a sub-block carrying a shrinking share of the weight, measured by exhaustive enumeration at $6.11\%, 2.74\%, 1.23\%$ for $K = 8, 10, 12$ (and $5.20\%, 2.31\%, 1.03\%$ of the patterns). Injectivity and the surjectivity onto that block are exact, and were verified by enumeration at the same K with zero duplicates; what is not true is that the sum runs over consecutive integers, and the shrinking share above is how far from it the truth is. In the rescaled variable $u = m/2^K$ the window sum has a clean limit. Since $L/P = 1/(3 \cdot 2^K)$, both Lm/P and $k_0 m/P$ equal $u/3$, so $D_W(m) \sim (3L/2\pi i u)\big(e(2u/3) - e(u/3)\big)$ and

$$\frac{\operatorname{Re} D_W(m)}{L} \;\longrightarrow\; g(u) := \frac{3}{\pi u}\cos(\pi u)\sin\Big(\frac{\pi u}{3}\Big),$$

which matches the exact sum to $2 \cdot 10^{-9}$ at $m = 1$ and to $5 \cdot 10^{-3}$ at $m = 2000$ for $K = 10$. The kernel has mean zero, $\int_0^\infty g = 0$ to nine digits, because it is a difference of two sinc profiles and the window omits the origin; numerically $\sum_m g(m/2^K) = -1.87$ against $\sum_m |g(m/2^K)| = 2147$. What that zero mean gives is an annihilation, not yet an explanation: every locally polynomial model of the weight c contributes nothing, so the leading term of any smooth model dies. Whether it also accounts for the *size* of Err is open and the evidence points both ways. An independent recomputation puts the smooth-model consequence at 2–4% of Err ($-0.0113, -0.0135, -0.0103$ against $-0.3066, -0.3893, -0.4988$ at $K = 8, 10, 12$), which would leave the value carried by the fine correlation between c and the kernel. Smoothing c over 16 consecutive frequencies inside the block instead reproduces the block sum to within 6% at $K = 8$ and 0.5% at $K = 12$, and overshoots as the smoothing widens. The two computations run over different index sets, so they do not yet contradict each other; until that is settled the mechanism should be stated as the annihilation and nothing more. The sharp form of what remains is $\sum_{m\ge 1} c(m)g(m/2^K) = O(1)$ against $\sum_m c(m)|g(m/2^K)|$ growing geometrically: a local limit theorem for the weight against one fixed mean-zero kernel.

Computing both directly, with c read off by the digit algorithm, gives for $K = 4, \dots, 12$ the signed values $-0.880, -0.930, -0.936, -0.955, -0.955, -0.948, -0.942, -0.930, -0.915$ against absolute values $6.0, 9.3, 14.0, 20.8, 30.5, 44.3, 63.7, 91.1, 129.5$. The signed sum is the same flat -0.93 found earlier in the sign-pattern coordinates, reached here through a completely different route, while the absolute sum grows like 1.4^K. Those signed values are Err/L, not Err; the deviation is comparable to the main term and cancels most of it, and it is worth saying so plainly, because read as $\mathrm{Err} = O(1)$ they would settle the problem on the spot.

How much cancellation is needed, exactly. What suffices is $\mathrm{Err} \le CL$: then $\sum_{k \in W} F_K(k) \le (1 + C)2^{-(K+1)}L$, and with $L = R = (3/2)^K$ this is $E(R) \ll R^{-0.7095}$, summable over octaves with room to spare. So the whole problem is an upper bound on one deviation by a constant multiple of the main term. The triangle inequality does not give it, and the margin can be measured. Writing $D(\nu) = \sum_{k \in W} e(k\nu/P)$ exactly and summing $|c_\varepsilon D(\nu_\varepsilon)|$ over all 3^{K+1} patterns gives 10.6, 24.6, 50.5, 105.2 times L at $K = 4, 6, 8, 10$, roughly doubling every two steps, against the true $\mathrm{Err}/L = -0.868, -0.936, -0.955, -0.943$ on the same windows. The identity itself is exact: $2^{-(K+1)}(L + \mathrm{Err})$ reproduces the direct sum $\sum_{k \in W} F_K(k)$ to every printed digit at each of those K. The phase in D is the whole content — dropping it and keeping only $|\sin(\pi L\nu/P)/\sin(\pi\nu/P)|$ turns a quantity near $-0.94L$ into one near $+40L$. Restricting to the consecutive block flatters the bound but does not rescue it: $6.96, 13.4, 24.1, 43.8$ times L at the same K.

The second moment is exact, and it names the obstruction again. Average over the window position instead. Writing $D_{k_0}(\nu) = e(k_0\nu/P)G(\nu)$ with $G(\nu) = \sum_{j<L} e(j\nu/P)$, every $\nu \neq 0$ is killed by the average, so $\mathbb{E}_{k_0}[\mathrm{Err}] = 0$ exactly. For the second moment, distinct ε give distinct ν_ε with $|\nu_\varepsilon| \le 3^{K+1} - 2^{K+1} < P$, so no two collide modulo P and

$$\mathrm{Var}_{k_0}(\mathrm{Err}) = \sum_{\varepsilon \neq 0} c_\varepsilon^2 |G(\nu_\varepsilon)|^2.$$

The no-collision step is exactly Lemma 1, not the range bound: differences $\nu_\varepsilon - \nu_{\varepsilon'}$ have coefficients in $\{0, \pm 1, \pm 2\}$ and can exceed P in absolute value, so the range alone does not close it. What closes it is that 3^{K+1} divides such a sum only if every coefficient vanishes, which is the same 3-adic induction, with the same $\{0, \pm 1, \pm 2\}$ coefficient set the lemma was stated for.

This reproduces the variance computed directly from every window in a full period, to six digits at $K = 6, 8, 10, 12, 14$. Now $\sum_\varepsilon c_\varepsilon^2 = \prod_i (1 + 2 \cdot \frac{1}{4}) = (3/2)^{K+1}$ exactly, so the normalised weight makes the coordinates independent with $\mathbb{P}(0) = \frac{2}{3}$ and $\mathbb{P}(\pm 1) = \frac{1}{6}$. Solving $p(1-p) = \frac{1}{6}$ gives $p = \frac{1}{2}(1 + 3^{-1/2}) = 0.7886751$, and then $1 - 2p(1-p) = \frac{2}{3}$: the auxiliary digit law is exactly $\eta = B - B'$ for B Bernoulli(p), verified to $3 \cdot 10^{-17}$ against the product law at $K = 4, 6, 8$.

So the second moment is governed by the correlation integral of the *biased* Bernoulli convolution at the same ratio $2/3$ with $p = 0.78868$. It is tempting to call that a harder instance, on the ground that the dimension ratio $H(p)/\log(3/2) = 1.2719$ is below the unbiased $\log 2/\log(3/2) = 1.7095$. That compares the wrong exponent. The quantity governing an L^2 statement is $\log\big(p^2 + (1-p)^2\big)/\log(2/3)$, and at this p the numerator is $\log \frac{2}{3}$ exactly, so the exponent is 1 — the auxiliary measure is exactly L^2-critical, which is the same fact as $\alpha(p) = 0$ below, not a harder open problem. For the unbiased measure the two exponents coincide, both 1.7095, which is what makes the substitution easy to get wrong. The consequence

is favourable rather than otherwise: a variance linear in K gives a typical Err of order $L\sqrt{K}$, and $R^{-0.70951}\sqrt{\log R}$ is still summable over octaves. Numerically the auxiliary object is well behaved — $\mathrm{Var}(\mathrm{Err}/L)$ measures $1.90, 2.15, 2.54, 2.91, 3.28$ at $K = 6, 8, 10, 12, 14$, linear in K and so logarithmic in R, which is exactly why the quantiles of Err/L over positions are stable — but the proof one would need is a harder case of the theorem being proved.

That logarithm is not an accident, and saying why sharpens the whole picture. For the biased measure the full-period average of the truncated square is $(1 - 2p(1-p))^{K+1}$, so the main term of the octave sum has exponent

$$\alpha(p) = 1 + \frac{\log\bigl(1 - 2p(1-p)\bigr)}{\log(3/2)},$$

which is negative exactly when $p(1-p) > \frac{1}{6}$, that is for p strictly between the two roots $\frac{1}{2}(1 \mp 3^{-1/2}) = 0.2113249$ and 0.7886751. But $p(1-p) = \frac{1}{6}$ is precisely the condition that defined the auxiliary law: the off-diagonal weight of the normalised c_ε^2. So the auxiliary measure sits exactly on the boundary $\alpha = 0$, and a critical exponent produces a logarithm rather than a power. That is an identity, not a numerical coincidence, and it accounts for the measured linear growth in K. Measured against the prediction, the octave slope is -0.783 at $p = \frac{1}{2}$ (against -0.7095), -0.364 at $p = 0.3$ (against -0.3435), -0.007 at the critical p (against 0), $+0.043$ at $p = 0.2$ (against $+0.0488$) and $+0.509$ at $p = 0.1$ (against $+0.5106$). At $p = \frac{1}{2}$ the measured decay beats the main term by about 0.07, which is the deviation helping rather than hurting, consistent with $\mathrm{Err}/L \approx -0.94$.

Two thresholds in p therefore sit near each other and should not be confused: the measure is forced singular below $p = 0.14028$, where $H(p) < \log(3/2)$, while the main term stops supplying decay already at $p = 0.2113$. In between, the measure has dimension one and the elementary route gives nothing — the same predicament as at $2/3$ itself, but with less room. There is in fact a third threshold between them, and naming it matters for what follows. The biased factor is $|(1-p) + pe^{i\varphi}|^2 = 1 - 2p(1-p)(1-\cos\varphi)$, whose arithmetic average over an equidistributed phase is $1 - 2p(1-p)$ — the quantity $\alpha(p)$ above is built from — while its logarithmic average is $2\log(1-p)$, by $\int \log|a + be^{i\varphi}|^2 = 2\log\max(a,b)$. Annealed and quenched averages therefore give different thresholds: the typical per-frequency decay turns over at $(3/2)(1-p)^2 = 1$, that is $p = 0.183503$, strictly between the entropy threshold 0.140277 and the main-term threshold 0.211325. At the latter the annealed log-factor is -0.40542 against $\log(2/3) = -0.40547$, which is $\alpha = 0$ to five digits.

This is worth stating because the biased coin is also the one available second control. Every statistic in this note was calibrated at $1/\varphi$ and $2^{-1/2}$, both algebraic with the singular one Pisot, and the Pisot reading is forced rather than measured. The biased coin at the *same* ratio $3/2$ supplies a structurally different singularity — entropy deficiency rather than arithmetic coincidence — with identical frequencies, lattice and window, and its singularity needs nothing deep: the dimension is at most $H(p)/\log(3/2) < 1$ for $p < 0.140277$, and a measure on the line of dimension below one is not absolutely continuous. But it can only be used outside the middle band. Below 0.140277 the measure is singular and a criterion should fire; above 0.211325 the main term supplies decay and it should pass; between them the measure has dimension one while the main term already diverges, so neither outcome carries information. A control point placed there has no known answer to calibrate against.

What such a control could establish is detection of entropy-deficient singularity at a non-Pisot ratio, which is more than this note currently has. What it cannot supply is a measure

of dimension one that is singular for an arithmetic reason — which is what μ at the unbiased $2/3$ would have to be, and for which no example is known. The hole therefore narrows rather than closes. That dimension claim deserves its warrant, since the note leans on it more than once. At $\lambda = 2/3$ the digit map is injective, and elementarily so: if $\sum_{i \le n} a_i (2/3)^i = 0$ with $a_i \in \{-1, 0, 1\}$, clearing denominators gives $\sum_i a_i 2^i 3^{n-i} = 0$, and reducing modulo 3 leaves $a_n 2^n \equiv 0$, so $a_n = 0$ because $a_n \in \{-1, 0, 1\}$; induction removes the rest. Hence $2/3$ is a root of no nonzero polynomial with coefficients in $\{-1, 0, 1\}$, distinct digit strings stay separated at every level, and exponential separation holds trivially rather than by any Diophantine input. Under that hypothesis the dimension is $\min\{1, H(p)/\log(3/2)\}$, equal to one exactly when $H(p) \ge \log(3/2)$. The arithmetic just given is elementary and was checked here; the theorem it feeds is quoted under the same caveat as the criteria above, from the literature rather than from a reading of its proof. The point of stating it is the contrast: dimension at $2/3$ is settled, and settled by an injectivity that costs nothing, while absolute continuity is open — so the whole difficulty of this note lives in the gap between full dimension and an L^2 density, not in any scarcity of separation.

Applying the substitution to the reduction itself is a check worth making, since a formulation that quietly hides the conclusion in its error term would pass every other test here. Measuring Err/L at the octave position for $K = 6, \dots, 16$ gives -0.94 at $p = \frac{1}{2}$, -0.78 at $p = 0.35$, -0.38 at the critical p, -0.12 at $p = 0.10$ and -0.044 at $p = 0.05$, each stable in K to the third digit. It is bounded, and comfortably so, at every p including the singular ones. The deviation term therefore carries no part of the conclusion: all the decay lives in $\alpha(p)$, and $\mathrm{Err} = O(L)$ is exactly the supplementary fact it was advertised as. That is the outcome one wants, and it was not guaranteed.

That the bound cannot be uniform in the position is worth recording separately, since it rules out every argument that does not look at where the octave sits. Sweeping k_0 over a full period, the median of Err/L is -0.34, the ninetieth percentile $+1.19$ and the ninety-ninth $+6.4$, all stable in K; but the maximum is $13.0, 22.7, 41.6, 75.0, 134.4$ at $K = 6, \dots, 14$, growing like $(4/3)^K$. The extreme positions are the windows meeting the peak at the origin, where $F_K = 1$ and $\mathrm{Err} \approx 2^{K+1} - L$. About 11% of positions have $\mathrm{Err} > L$ at every K tried. The octave position $k_0 = R$ sits near the bottom of the distribution at -0.94, and a proof has to know that.

What the octave position does to the phases. It does something exact, and it is the cleanest description of the mechanism we have. The terms that matter are those with $|\nu| \lesssim P/L = 3 \cdot 2^K$, since G is small outside that range — which is why a block carrying about one percent of the weight carries the sum. On that range the phase is $e(k_0 \nu / P)$ with $k_0 = R = (3/2)^K$, and

$$\frac{k_0 \nu}{P} = \frac{(3/2)^K \nu}{3^{K+1}} = \frac{\nu}{3 \cdot 2^K} \qquad \text{because} \qquad (3/2)^K \cdot 2^K = 3^K.$$

The phase is exactly linear in ν, with slope the reciprocal of the effective range, so it completes one turn on each side of the origin across precisely the region where G lives. In the rescaled variable $u = \nu / 2^K$ it is the single frequency $e(u/3)$. Since k_0 must be an integer the true slope is $\lfloor (3/2)^K \rceil / 3^{K+1}$, but the rounding is immaterial: the ratio to the idealised slope is 1.00024 by $K = 14$, and substituting the idealised phase reproduces Err/L to five digits (-0.93644 against -0.93566 at $K = 6$, -0.91563 against -0.91559 at $K = 12$).

Now g is the normalised transform of the octave band, which in this variable is $[\frac{1}{3}, \frac{2}{3}]$, so $\hat{g}$ is a multiple of $1_{[1/3,2/3]}(|t|)$. The frequency the octave supplies is exactly $\frac{1}{3}$: the left endpoint of that band. Evaluating $\hat{g}$ by tapered quadrature gives $-0.00000, -0.00000$ at $t = 0.20, 0.30$ below the band, 1.50000 at $t = 0.40$ and $t = 0.50$ inside it, -0.00000 at $t = 0.80$ above it, and 0.75000 at both endpoints $t = \frac{1}{3}$ and $t = \frac{2}{3}$ — exactly half the interior value, as a jump discontinuity gives. So the octave, which is precisely the window whose starting point equals its length, places the phase frequency on the edge of its own band, where the transform takes half its interior value rather than the full value it would take strictly inside or the zero it would take strictly outside. That is a clean account of why the answer is a bounded $O(1)$ number instead of either extreme, and it is specific to $k_0 = R$. It is not a proof: the leading term is only the smooth part of the weight, and the deviation of c from smooth is exactly what carries the value. But it says what is special about the octave, in one line, exactly.

The octave sits between two exact zeros, and that proves most of the cancellation. The same fact has a second face, in the original variable, and this one is an identity with no numerics in it. Read $F_K(x) = \prod_{i=0}^{K} \cos^2(\pi x 2^i / 3^{i+1})$ as a function on the reals. At $x = R = (3/2)^K$ the factor $i = K - 1$ has argument

$$\pi \cdot \frac{(3/2)^K \, 2^{K-1}}{3^K} = \pi \cdot \frac{2^{K-1}}{2^K} = \frac{\pi}{2},$$

so it vanishes; at $x = 2R$ the factor $i = K - 2$ has argument $3\pi/2$ and vanishes. The octave $[R, 2R]$ is therefore *exactly* the gap between two consecutive zeros of F_K, for every K, identically and not approximately. Confirmed in sixty-digit arithmetic: the identified factor is the smallest of the $K + 1$ at each endpoint and equals zero to working precision at $K = 6, 10, 14$.

The consequence is the strongest structural statement here. Since $k_0 \nu / P = \nu / (3 \cdot 2^K)$ the phase is a character modulo $3 \cdot 2^K$, and its c-weighted law is exactly uniform: for every h with $v_2(h) \le K - 1$ the expectation $\mathbb{E}[e(h\theta)] = \prod_n \cos^2(\pi h 3^{n-1} / 2^n)$ vanishes, since the single index $n = v_2(h) + 1$ puts that argument at exactly $\pi/2$. So replacing $G(\nu)$ by $G(0) = L$ gives $\mathrm{Err}/L = -1$, against the measured -0.955: the phases alone annihilate the entire main term, and the bulk of the cancellation a proof needs is an identity rather than a numerical coincidence.

Three things should be said plainly about how much that buys, and the first of them takes most of it back. The bound $\mathrm{Err} \ge -L$ is free: F_K is a square, so $\sum_{k \in W} F_K \ge 0$, and the window identity turns that into $L + \mathrm{Err} \ge 0$ immediately — no Fourier expansion, no equidistribution, at every p and every window position. Measured, $\min_W F_K$ is $8.7 \cdot 10^{-13}$ at $K = 8$ falling to $7.0 \cdot 10^{-24}$ at $K = 20$, positive throughout. The zeroth-order computation lands exactly on that free floor: it predicts $A_K = 1 + \mathrm{Err}/L = 0$ where the truth is about 0.05, so it captures none of the quantity at issue and bounds it in neither direction. Against L the step looks like most of the cancellation; against A_K, which is what must be bounded, it is none of it.

Second, the zero is at the real point $(3/2)^K$, while k_0 must be an integer; at the integer start the zeroth-order term is small but not zero, measuring $2^{K+1} F_K(k_0) = 4.8 \cdot 10^{-5}, 5.5 \cdot 10^{-5}, 7.0 \cdot 10^{-7}, 2.3 \cdot 10^{-8}$ at $K = 6, 8, 10, 12$. It is shrinking, so nothing depends on this, but "exactly" belongs to the real point. Third, the character sum obeys $\sum_\varepsilon c_\varepsilon e(k\nu_\varepsilon / P) = \prod_i \big(1 + \cos(2\pi k n_i / P)\big) = 2^{K+1} F_K(k)$, since $1 + \cos 2\pi t = 2\cos^2 \pi t$. So $L + \mathrm{Err} = 2^{K+1} \sum_{k \in W} F_K(k)$ identically, and A_K is the normalised window sum itself — verified to eight digits by computing

it both ways at $K = 6, 8, 10, 12$. So A_K is the same object throughout, and what the exact zero contributes is a clean description of the octave's position rather than progress against it.

The integrality drops out, and what is left is a Riesz product. The one reformulation here that does change the shape of the problem is the passage to the continuum. Comparing A_K with

$$\int_1^2 \prod_{m=0}^{K} \Big(1 + \cos\big(2\pi x (3/2)^m/3\big)\Big)\, dx$$

is not an approximation but an identity of integrands. Substituting $k = Rx$,

$$\frac{R\,2^i}{3^{i+1}} = \frac{(3/2)^K 2^i}{3^{i+1}} = \frac{1}{3}\Big(\frac{3}{2}\Big)^{K-i},$$

exactly, so $\prod_i\big(1 + \cos(2\pi k 2^i/3^{i+1})\big)$ and $\prod_m\big(1 + \cos(2\pi x(3/2)^m/3)\big)$ are the same function with the index reversed. A_K and the integral are therefore the Riemann sum and the integral of one function, and only quadrature error separates them. Measured, that error is $A_K - I_K = 0.342003 \cdot 2^{-(K+1)}$, with the constant flat to six digits across $K = 9, \ldots, 27$: it halves at every step, and the earlier impression that the agreement was erratic — gaps of $1.7 \cdot 10^{-5}$ at $K = 8$ against $1.2 \cdot 10^{-4}$ at $K = 12$ — was noise in an under-resolved quadrature, not a feature. The true gaps are $6.7 \cdot 10^{-4}$ and $4.2 \cdot 10^{-5}$, on the clean geometric ladder. So Err $= O(L)$, and therefore the whole L^2 target, follows from the boundedness in K of that integral: a Riesz product with the non-Pisot ratio $3/2$, integrated over a single octave. The implication is one way only, and the sharper statement is worth having. Writing I_K for that integral and B_K for the mass of $Q^2 dv$ on the block $[(3/2)^K, (3/2)^{K+1})$, the substitution $\theta = (2/3)^K v$ relates them, but not by a clean identity: it carries the weight $\prod_{i\geq 1} \cos^2(\pi(2/3)^i v)$, which is bounded above and below on the block and is the reason the comparison is not exact. Writing $C_K = (3/4)^K I_K/(3B_K)$, the measured values run from 119.0 at $K = 8$ to 244.3 at $K = 12$, with median 187.1 over $K = 8, \ldots, 35$, a level shift near $K = 11$, and flat at 0.9968 per step over $K = 13, \ldots, 35$: bounded within a factor of two, which is all that is needed, and not a constant. Two earlier drafts got this wrong in opposite directions — one quoted the ratio at two endpoints as though it were constant, the second promoted it to an exact identity $B_K = (3/4)^K I_K/2$, which would force $C_K \equiv 2$ and is refuted by the measurements above. With the comparison in hand,

$$\mu \in L^2 \iff \sum_K (3/4)^K I_K < \infty,$$

so I_K may grow, just not at rate $4/3$; boundedness of I_K is sufficient and strictly stronger, since $B_K = 0.9^K$ is summable while $B_K(4/3)^K = 1.2^K$ diverges.

What the measurements exclude should be stated without choosing an estimator. Over $K = 8, \ldots, 35$ the per-step rate of I_K is 1.0032 as an arithmetic mean of the ratios, 0.9973 as a geometric mean, and 0.9829 by least squares on $\log I_K$; the spread is Jensen's gap, $\sigma^2/2 = 0.0061$, and the sign of the drift is not determined by the data. No single rate should be quoted, and earlier drafts quoted two different ones. What survives every estimator is the exclusion that matters: $\log(4/3)$ sits 13.6 standard errors above the mean increment and would have produced a factor of 2362 across the window, against an observed total factor of 0.930. So the L^2 failure mode is dead over the reachable range. The resolution is model-dependent and both

readings belong here: treating the increments as a random walk gives a two-standard-error band of $[0.956, 1.040]$ on the per-step rate, while under the bounded hypothesis actually being tested the endpoint estimator resolves a total log-drift of $0.7506/27 = 0.0278$. Boundedness itself — the strictly stronger claim — is untested either way, and every sub-exponential law stays invisible: $K^{1/2}$ and $(\log K)^b$ for $b < 2.11$ are inside the noise, while K^1 would not be.

The direction the theorem needs is close to rigorous. Since H vanishes at both endpoints of the window, $|A_K - I_K| \le (2R)^{-1} \int_W |H'| \le \pi D_K I_K$ with $D_K = \sum_i \theta_i \big(\int H_i / \int H \big)$ measured at 1.10 ± 0.03 for $K = 12, \dots, 20$. So $\sup_K I_K < \infty$ would give $\sup_K A_K < \infty$ and hence $\mu \in L^2$, once $D_K = O(1)$ is proved; the converse remains measured only. No property of $(3/2)^K$ modulo 3^{K+1} is available or needed, and the arithmetic of the lattice, the window position and the choice of integer start have all disappeared. That is the classical form of the question — Erdős's 1939 setting, and Kahane–Salem territory — reached from the entropy side.

Why the Riesz-product literature does not reach it, and it is not the ratio. Two thresholds are usually run together. Ratio ≥ 2 makes every $\{0, \pm 1\}$ combination of the frequencies nonzero, so the product has mean one; ratio ≥ 3 makes those combinations distinct, so the coefficients are read off uniquely. Both hold here. The frequencies $\lambda_m = (3/2)^m/3 = 3^{m-1}/2^m$ have strictly decreasing 2-adic valuations, so $\sum_m \varepsilon_m \lambda_m = 0$ with $\varepsilon \in \{0, \pm 1\}^{K+1}$ forces ε_K even, hence zero, hence all zero by induction — the same argument as Lemma 1 with the roles of 2 and 3 exchanged. The set is dissociate for the coefficients Lemma eflem:dis allows, and the constant term of the product is exactly 1. It is not dissociate beyond them: $3n_m = 2n_{m+1}$ identically for $n_m = 3^m 2^{K-m}$, so relations with a coefficient 3 exist in quantity, and an earlier phrasing here calling the set dissociate in the strongest sense was wrong.

What fails is not that the frequencies leave $\mathbb{Z}$ — scaling by $3 \cdot 2^K$ returns the integers $n_m = 3^m 2^{K-m}$ with ratio $3/2$, a circle Riesz product of the classical kind. It is that the frequency set depends on K: each new factor rescales all the others, so there is no one sequence for the classical theory to attach to. Since $\lambda_0 = \frac{1}{3}$ the product has period $3 \cdot 2^K$, and A_K is its average over a window that is $1/(3 \cdot 2^K)$ of that period. The classical theorems conclude about the mean over the whole group, or about singularity and dimension; none of them bounds a partial-period average. That last clause is a survey-wide negative, assembled from a literature search covering Riesz, Zygmund, Peyrière, Brown–Moran, Kilmer–Saeki, Parreau, Ritter and Fan, and I have not read those sources myself; the dissociateness and constant-term statements around it are elementary and were checked here in exact arithmetic, but the claim that nobody has bounded such a quantity should be treated as the current state of a search. On the Bohr compactification, where the classical criterion applies verbatim, it gives orthogonality to Haar measure, which says nothing about $[1, 2]$ because $\mathbb{R}$ is Haar-null there. So the gap is not lacunarity but locality.

Two measurements make the position concrete. Over one period at $K = 10$ the 3072 unit windows have mean exactly 1, median 0.804 and maximum 21.4 at the alignment spike $t = 0$ — so, taking the maximum over unit windows of the window *mean* of P_K in the x coordinate, $\sup_t \int_t^{t+1} P_K = \|f\|_2^2 (4/3)^K$, measured $1.20646, 1.20618, 1.20621$ at $K = 8, 10, 12$ against $\|f\|_2^2 = 1.206310$. A proposed correction putting that supremum at $2\|f\|_2^2 (4/3)^K$ does not survive a direct measurement in this normalisation; the factor two presumably belongs to a different window length — the window $[1, 2]$ ranks 486th, at the sixteenth percentile. That rank is one value of K and it drifts by eight to twenty per cent over $K = 3, \dots, 12$,

so it locates the octave in the lower part of the distribution without pinning it. The octave is also the immediate neighbour of the one growing window: the ratio of the window at the origin to the next is $32, 107, 269, 372, 411, 451$ at $K = 4, \dots, 12$, itself of order $(4/3)^K$. And the independence prediction $\prod_m \int_1^2 \big(1 + \cos 2\pi x \lambda_m\big) dx$ converges to 0.2784, about five times A_K, with the the shortfall being genuine negative correlation between the factors. Whether it accumulates cannot be read off this ratio: the denominator converges by $m \approx 10$, so the ratio simply tracks A_K, and inherits both the decline through the transient below $K \approx 25$ and the flatness after it. Readings of "stable" and of "monotonically accumulating" have both been offered here, and each was measured over a range sitting mostly inside the transient.

The trough has a one-line cause, and it is a single factor. Of the $K + 1$ factors, only the lowest is small on $[1, 2]$:

$$\int_1^2 \Big(1 + \cos \frac{2\pi x}{3}\Big) dx = 1 - \frac{3\sqrt{3}}{2\pi} = 0.1730067,$$

against a mean of 1 over the whole line. It takes the value $\frac{1}{2}$ at both endpoints and vanishes at the midpoint $x = \frac{3}{2}$. Every other factor averages about one on $[1, 2]$ — measured $1.000, 1.212, 1.041, 1.154, 1.036, 1.017$ for $m = 1, \dots, 6$ — so the independence value 0.2784 is essentially that one integral times a product of terms near unity. The endpoints are themselves zeros of low-index factors: $x = 1$ annihilates the $m = 1$ factor and $x = 2$ the $m = 2$ factor, which is the continuum face of the exact zeros of F_K at R and $2R$.

So the anatomy of $A_K \approx 0.05$ is: a factor 0.173 from the lowest frequency alone, a factor near 1.6 from the rest, and a further factor near 0.18 from correlation among the factors. The first two are explicit and bounded. The whole of the difficulty is the third.

That splitting can be made exact, and it turns the target into a correlation inequality with a normaliser one can write down. The product of the individual means converges absolutely: each factor has

$$\Big| \int_1^2 \big(1 + \cos 2\pi x \lambda_m\big) dx - 1 \Big| = \frac{|\sin 4\pi\lambda_m - \sin 2\pi\lambda_m|}{2\pi\lambda_m} \le \frac{1}{\pi\lambda_m} = \frac{3}{\pi(3/2)^m},$$

which is summable, so $\prod_m \int_1^2 (1 + \cos 2\pi x \lambda_m) dx$ converges; numerically it is 0.2783515, already stable to seven digits by $m = 30$. Since that limit is a fixed positive number, $A_K = O(1)$ is exactly the statement that the ratio

$$\frac{\int_1^2 \prod_{m \le K} \big(1 + \cos 2\pi x \lambda_m\big) dx}{\prod_{m \le K} \int_1^2 \big(1 + \cos 2\pi x \lambda_m\big) dx}$$

stays bounded as factors are added — measured near 0.18 and flat in K. This is a restatement rather than a bound, but it is the sharpest one available: it names the difficulty as the failure of the factors to behave independently on a fixed interval, and it supplies the normaliser explicitly, so nothing in the problem now depends on an unevaluated constant.

Where that dependence comes from is visible at the start of the product. Taking the factors in order, the ratio above reads 1.0000, 1.0000, 0.2652, 0.3297, 0.4752, 0.2683, 0.2394, 0.1623 for the first one through eight. The first two are *exactly* uncorrelated; the defect is created when the third joins, and the reason is a coincidence of zeros. Inside $[1, 2]$ the lowest factors each vanish once, and at different places: $m = 0$ at the midpoint $x = \frac{3}{2}$, $m = 1$ at the left endpoint

$x = 1$, $m = 2$ at the right endpoint $x = 2$, then $m = 3$ at $\frac{4}{3}$ and $m = 4$ at 1.4815. Three zeros at the two ends and the middle leave the product small across the whole interval, which no independence estimate can see, since each factor separately is small only near its own zero. After that the ratio fluctuates rather than decaying, later factors contributing correlation and anti-correlation in turn.

Carried to 26 factors the ratio stays in $[0.1596, 0.4752]$, with the maximum attained early, at $n = 4$. Whether it is settling has to be measured carefully: the range over $n \geq n_0$ shrinks as n_0 grows for purely formal reasons, so it is no evidence at all. Fixed-width blocks are not forced to narrow, and over n = 2–6, 7–11, 12–16, 17–21, 22–26 the block ranges are $0.236, 0.088, 0.091, 0.039, 0.036$ with means $0.316, 0.192, 0.239, 0.180, 0.174$. Read alone that looks like damping toward 0.175. It should not be read alone: the same quantity carried to $K = 40$ has its five-point running mean fall to 0.041 and then rise again over the last eight values, so the apparent settling is inside a window that stops before the next excursion. Taken with the limitation above — the oscillation and a logarithm are the same size over every K that can be computed — the honest summary is that the ratio has stayed between about 0.16 and 0.48 for forty factors and that nothing in the data decides whether it stays bounded.

One more feature of the parameter deserves recording, because it explains the silence in the literature better than anything else here. At bias p the same product is $\prod_m (1 + a \cos 2\pi x \lambda_m)$ with $a = 2p(1-p)/(1-2p(1-p))$, and $a = 1$ exactly at $p = \frac{1}{2}$. The one classical extension below ratio 3 requires $\sup_j |a_j| < 1$ strictly. It therefore covers every bias except the fair coin, and degenerates precisely at the case that is open. The endpoint $a = 1$ is where this machinery has always stopped.

What is left is exact,

$$L\,A_K = \sum_{\varepsilon} c_\varepsilon\, e(\theta_\varepsilon)\big(G(\nu_\varepsilon) - L\big),$$

a decorrelation between the now exactly uniform fractional part of θ and its own envelope. The substitution kills this cleanly, as it must: at bias p the vanishing factor becomes $1 - 4p(1-p)$, zero only at $p = \frac{1}{2}$. The mechanism belongs to the fair coin, not to the functional equation, and supplies nothing where the measure is singular. Independently Err/L has been carried to $K = 38$, a window of 4 914 370 terms, where it reads -0.954.

In these terms the remaining task is compact. Writing $A_K = 1 + \mathrm{Err}/L$, the window identity gives $E(R) \leq 2^{-(K+1)} L A_K = \frac{1}{2} R^{-0.7095} A_K$, so $A_K = O(1)$ is the whole of what is left, and the zeroth-order computation above says A_K has no main term at all. Measured at the octave for every K from 6 to 40 — the last window has 11 057 332 terms — it runs between 0.0399 at $K = 32$ and 0.0846 at $K = 12$, a factor of 2.12, oscillating rather than settling. No trend should be read into it: log-linear fits give $R^{-0.030}$ from $K \geq 6$, $R^{-0.039}$ from $K \geq 10$, $R^{-0.011}$ from $K \geq 20$ and $R^{+0.014}$ from $K \geq 26$, the sign turning with the fitting window, and the five-point running mean falls from 0.069 to 0.041 and then rises again over the last eight values. An earlier draft of this note fitted $R^{-0.047}$ on a stride of four and read it as slow decay; that was the stride's trend, not the sequence's.

Carried further, on an independent code path with composite Gauss–Legendre quadrature and exact integer phase arithmetic, the integral reaches $K = 37$ and settles: over $K = 25, \ldots, 37$ the log-drift is $+0.00009$ per step at level 0.0427, and the apparent decay in the earlier ladder is a transient below $K = 25$. Two structural facts come with it, both of which change what a proof should look for. The recursion is exact, $I_K = I_{K-1} + C_K$ with $C_K = \int_1^2 \cos(2\pi f_K t) P_{K-1}(t)\, dt$ the Fourier coefficient of the previous product at its own next

frequency — and its multiplier is one, not less: the geometric mean of I_K/I_{K-1} over $K \geq 20$ is 0.9990 ± 0.017. *So the theorem cannot come from a contraction*, and that route is closed by measurement rather than by argument. What the same data does show is mean reversion: $\mathrm{Var}(\log I_{K+j} - \log I_K)$ saturates near 0.019 instead of growing linearly in j, the lag-8 variance ratio being 0.28. Boundedness here is a fluctuation statement about a driftless positive sequence, not a decay statement.

Which rules out the obvious way to use it. The per-step standard deviation is 0.071, so independent increments would give $\mathrm{Var} = 0.0403$ over eight steps against the 0.0113 implied by the measured ratio; each step contributes about 0.005 to a sum that therefore diverges linearly. So $\sum_K C_K^2$ does *not* converge, and a proof cannot come from the increments being square-summable — the natural martingale route, and it is closed by the same data that suggests it. Bounded variance over j steps is bought entirely by cancellation between increments. What has to be proved is that the C_K are negatively dependent, a mixing statement, and not that they are small.

That target can be localised. Writing $\Phi(u) = \int_1^2 \cos(2\pi u t) P_{K-1}(t)\,dt$ and using $f_{K+1} = \frac{3}{2} f_K$, the product-to-sum identity gives exactly

$$C_K = \Phi(f_K), \qquad C_{K+1} = \Phi\big(\tfrac{3}{2} f_K\big) + \tfrac{1}{2}\Phi\big(\tfrac{1}{2} f_K\big) + \tfrac{1}{2}\Phi\big(\tfrac{5}{2} f_K\big),$$

all against the same P_{K-1}, verified to 10^{-15}. The last term is negligible, and for a structural reason rather than by accident: the largest frequency of P_{K-1} is $\sum_{m \leq K-1} (3/2)^m/3 = 2f_K - \frac{2}{3}$, so $\frac{1}{2} f_K$, f_K and $\frac{3}{2} f_K$ all lie inside the spectrum while $\frac{5}{2} f_K$ lies outside it. Measured, $|\Phi(\frac{5}{2} f_K)|$ relative to C_{K+1} falls from $2 \cdot 10^{-4}$ at $K = 8$ to 10^{-10} at $K = 16$. So the recursion is effectively two-term: successive increments are one function sampled at f_K and at its two multiplicative neighbours $\frac{1}{2} f_K$ and $\frac{3}{2} f_K$, and the half-frequency term is the larger of the two in four of the five K examined. Whether that produces the required anti-correlation is not settled by the five sign pairs available, three of which flip.

The reduction should not be mistaken for progress. Expanding Φ over the same characters gives $\Phi(u) = \frac{1}{2} \sum_\varepsilon c_\varepsilon \big[W(\nu_\varepsilon + u) + W(\nu_\varepsilon - u)\big]$ with $W(v) = \int_1^2 e(vt)\,dt$ and $|W(v)| \leq \min(1, 1/\pi|v|)$, so the triangle bound on an increment is again the weight sitting within $O(1)$ of $\pm u$ — the same count that defeats the triangle inequality for A_K itself. Measured, that bound is $21.3, 43.4, 87.0, 172.2$ at $K = 8, 10, 12, 14$ against true values $9.7 \cdot 10^{-5}$, $5.3 \cdot 10^{-3}$, $1.5 \cdot 10^{-2}$, $2.6 \cdot 10^{-3}$: overshoots of four to five orders of magnitude, growing. The two-term form is exact and it is a simplification of shape; the cancellation it needs is the cancellation we started with, in the same amount.

What the closed routes have in common. Every route closed in this note ends the same way: as a weighted count of characters near a distinguished frequency, with the true value exponentially below the count. The triangle bound on Err over all 3^{K+1} patterns, the block weight $(4/3)^K$, the test-function identity, the second moment, and the increment bound just given are five instances, with overshoots from a factor 10 to 10^5 and growing in K.

The common step is not the upper bound; it is *linearisation*. Each of those routes expands $P_K = \sum_\varepsilon c_\varepsilon e(\nu_\varepsilon t)$ and then bounds term by term, and that discards the cancellation which is the whole content. But P_K is a product of factors in $[0, 2]$ and so is nonnegative: a route that works with it directly — covering $[1, 2]$ and bounding the function on each piece — discards nothing, because for a nonnegative integrand there is no cancellation to lose. That is exactly the difference between the routes above and the two that remain open. Counting the peaks

bounds $\sum$ height $\times$ width for the nonnegative P_K; the Erdős–Kahane argument is a digit count over $k \in [R, 2R)$ on the object itself. Neither linearises. That is not evidence either will work, and the peak count walks into $(3/2)^m$ modulo one; but it says which question to ask of a proposed route, and the question is whether it expands into characters, not whether it ends in an inequality.

The integral is a Birkhoff average for the $\times 3/2$ map. There is a second change of coordinates, and it is exact. Since $\theta_m(x) = x(3/2)^m/3 \bmod 1$ satisfies $\theta_{m+1} = \frac{3}{2}\theta_m \bmod 1$, and $1 + \cos(2\pi x(3/2)^m/3) = e^{\varphi(\theta_m)}$ with $\varphi(\theta) = \log(1 + \cos 2\pi\theta)$, substituting $x = 3\theta$ gives

$$I_K = 3\int_{1/3}^{2/3} \exp\Big(\sum_{m=0}^{K} \varphi\big(T^m\theta\big)\Big)\,d\theta, \qquad T\theta = \tfrac{3}{2}\theta \bmod 1. \tag{8}$$

Verified against the direct product to all printed digits at $K = 6, \ldots, 16$. So the question is whether a Laplace average of Birkhoff-type sums along the sequence $(3/2)^m$ modulo one, taken over Lebesgue measure on a fixed subinterval, grows at rate exactly one. That is thermodynamic formalism rather than counting: the natural objects are the transfer operator of T tilted by φ and its leading eigenvalue, and the natural statement is that the growth rate vanishes with a bounded prefactor. It reaches the $\times 3/2$ map, as everything here eventually does, but through pressure rather than through the orbit of a point — which is the one door the peak count could not use.

A correction belongs here, because the dynamical reading of (8) was wrong and the error survived several pages of this note. The identity itself is a change of variables and is correct; it was checked against the direct product to every printed digit. But the sequence θ_m is *not* the orbit of $\theta \mapsto \frac{3}{2}\theta \bmod 1$. Writing $\theta_m = u_m - k_m$ with k_m the stripped integer,

$$\theta_{m+1} = \tfrac{3}{2}(\theta_m + k_m) \bmod 1 = \tfrac{3}{2}\theta_m + \tfrac{1}{2}[k_m \text{ odd}] \bmod 1,$$

and the parity of k_m is not a function of θ_m. In exact rational arithmetic 212 of 429 consecutive pairs fail the plain map and none fails the skewed form; $x = 3$ gives $\theta = 0, \frac{1}{2}, \frac{1}{4}, \frac{3}{8}$, where the plain map sends 0 to 0. Nor is this repaired by another modulus — reducing modulo 2 fails as well, and multiplication by $3/2$ is an endomorphism of $\mathbb{R}/P\mathbb{Z}$ only if $\frac{3}{2}k \in \mathbb{Z}$ for every integer k, which holds for no P. The sequence is the orbit of no circle map at all; its natural home is a skew product carrying a parity bit, or the 2-adic solenoid. That is precisely the difficulty in Mahler's problem, and it is not something a one-dimensional transfer operator reaches.

So what (8) delivers is a Birkhoff-*type* sum along the sequence $x(3/2)^m \bmod 1$, and not a thermodynamic formalism with an operator behind it. The operator written below was built for a map that is not the dynamics here, which is the honest explanation of why it returned a degenerate eigenmeasure concentrated at a fixed point rather than anything about μ: its leading eigenvalue $4/3$ is a fact about that operator, and the $(4/3)^K$ in this problem is established independently, by the elementary window decomposition. Everything derived from the closed-form product rather than from a map is untouched — the window constant, $\|f\|_2^2 = 6\int_0^\infty Q^2$, the tail renewal, the block criterion and the line count all stand, since none of them uses a map.

The way the error survived is worth a sentence. The substitution was verified by checking the *integrand* in closed form, which is exactly the test that cannot see a false recursion; the recursion itself was asserted from a one-line manipulation and never checked. It surfaced only

because a separate line of work arrived at a parity-skewed coding and said plainly that it had not proved its coding equivalent to the one used here.

The atom was a frozen bit, and the annealed rate is exactly one. The correction above says the phase is not driven by a circle map; what drives it is $x_{m+1} = \{\frac{3}{2}x_m + \frac{1}{2}b_m\}$ with b_m the parity of $\lfloor (3/2)^m\theta \rfloor$, a skew product over the two-point fibre. That makes the earlier atom transparent. For $\theta < (2/3)^{K-1}$ every b_m is zero, which is the one deterministic branch fixing the origin, and it carries height 2^K on measure $(2/3)^K$ — the whole of the $(4/3)^K$. Every θ in the octave leaves that region in a single step and never returns. So the atom is an artefact of freezing the bit, and the object to look at is the operator averaged over the bits.

There the answer is exact and immediate. For every x,

$$\big(1 + \cos 2\pi\{\tfrac{3}{2}x\}\big) + \big(1 + \cos 2\pi\{\tfrac{3}{2}x + \tfrac{1}{2}\}\big) = 2 + \cos 3\pi x + \cos(3\pi x + \pi) = 2,$$

so averaging over b_m contributes exactly 1 at every step and the product telescopes: $\mathbb{E}_b\big[\prod_{m<K}(1+\cos 2\pi x_m)\big] = 1 + \cos 2\pi x_0$, for every x_0 and every K. The annealed octave integral is therefore the K-free constant $\int_{1/3}^{2/3}(1+\cos 2\pi\theta)d\theta = \frac{1}{3} - \frac{\sqrt{3}}{2\pi} = 0.0576689$, and the annealed growth rate is exactly one. Nothing is left to prove on that side.

What remains is the whole problem, and one exact rewriting is available. Since $x_m = \{\frac{3}{2}x_{m-1} + \frac{1}{2}b_{m-1}\}$,

$$\cos 2\pi x_m = \cos\big(3\pi x_{m-1} + \pi b_{m-1}\big) = (-1)^{b_{m-1}} \cos 3\pi x_{m-1},$$

so the weight is $w(x_m) = 1 + (-1)^{b_{m-1}} \cos 3\pi x_{m-1}$ identically. Writing J_K for the octave integral and $A = \frac{1}{3} - \frac{\sqrt{3}}{2\pi}$ for the annealed value, this expresses the increment $J_m - J_{m-1}$ as an integral of the past product against $(-1)^{b_{m-1}} \cos 3\pi x_{m-1}$, and hence $J_K/A = \prod_{m\le K}(1+\varepsilon_m)$ with ε_m that increment divided by J_{m-1}.

It must be said plainly that the product formula is a telescoping tautology: $\varepsilon_m = J_m/J_{m-1} - 1$ by construction, so verifying $\prod(1+\varepsilon_m)$ to 10^{-16} verifies $J_m = J_{m-1} + (J_m - J_{m-1})$, and an earlier draft here presented it as a result. Its content is the trig identity above, which gives ε_m an interpretation — a past-weighted correlation between the parity bit and the phase — exactly as ρ_n was given one earlier, and with exactly the same standing: a reformulation, not a reduction. For the same reason the target $\limsup K^{-1}\sum \log(1+\varepsilon_m) < \log\frac{4}{3}$ is the statement that J_K grows slower than $4/3$ per step, written in other coordinates.

The arithmetic behind the interpretation is real. With $N_m = (3/2)^m v = A_m + x_m$, the relation $N_{m-1} = \frac{2}{3}N_m$ gives $x_{m-1} = \{((2A_m \bmod 3) + 2x_m)/3\}$ — checked here in exact rational arithmetic, 780 of 780 — so x_{m-k} depends on x_m and $A_m \bmod 3^k$ alone while $b_m = A_m \bmod 2$. Since A_m ranges over about $L = (3/2)^m/3$ consecutive integers on the octave, the correlation of b_m with the k preceding phases is $O(3^k/L)$, small for k below $(m\log\frac{3}{2} - \log 6)/\log 3 = 0.36907\,m - 1.63$. That is a decay, not an exact vanishing, and the distinction matters: an independent check finds the inside-window floor falling like $1/L$ ($1.1\cdot 10^{-4}$ at $m = 24$ against $1.5\cdot 10^{-5}$ at $m = 30$) and the depth growing with m at the predicted slope, but no step at the boundary — a factor of 1.9 between the last inside and first outside value at $m = 24$, and at $m = 30$ none at all.

Nor does it license the conclusion an earlier draft drew from it. What is nearly orthogonal to b_m is the recent-k *truncation* of the product; $\prod_{j<m} w_j$ does not factor into a recent and a deep piece, so nothing transfers that smallness to ε_m itself. Measured, the truncated defect

inside the window is about 10^{-4} against a full $|\varepsilon_m|$ of 0.048: the part that is provably small is not the part that carries the gap.

As for the gap itself, it should not be called a constant. Computed exactly to $K = 36$ the ratio runs between 0.2307 and 0.4886, and its median moves from 0.2745 over $K = 8, \dots, 35$ to 0.2435 over $K = 25, \dots, 36$. On the octave $|\varepsilon_m|$ is flat across disjoint blocks, 0.0473 on $m = 25, \dots, 30$ against 0.0480 on $m = 31, \dots, 36$, so it does not decay where it matters; only an ensemble of generic octaves decays, at 1.4 standard errors, and this octave is a trough rather than a generic one. What is established is the absence of drift, -0.0121 per step against a critical $\log(4/3) = 0.28768$, at about eighteen standard errors — with the caveat that twelve increments of a single deterministic curve are not twelve independent observations, and that the estimate swings by a factor of ten as the window start moves from $K = 8$ to $K = 26$. Absence of drift is nevertheless what the theorem needs, since a driftless walk of step 0.058 gives $|\log r_K| \sim 0.058\sqrt{K}$.

The margin is wide and is worth stating correctly: the trivial bound $w \le 2$ gives $\log 2 = 0.693$ per step against an allowance of 0.28768, a factor of 2.41, while the measurement is -0.012. What is asked for is only an average bound — no decay rate, no summability, no convergence, no sharp constant. What is not available is any argument that supplies it, and the deciding measurement is pre-registered: run this octave alone to $K \approx 60$ in exact arithmetic and ask whether $|\varepsilon_m|$ on the disjoint blocks $m = 49, \dots, 54$ and $55, \dots, 60$ falls below 0.030. If it does, the ratio is a genuine constant; if it holds near 0.048, only the driftlessness is real.

The operator this suggests is explicit, and computing it says what the problem is. Preimages under T are $x = 2y/3$ always and $x = 2(y+1)/3$ only for $y < \frac{1}{2}$ — two branches below a half, one above, which is where the map's difficulty sits — so with weight $e^{\varphi} = 1 + \cos 2\pi x$ and $|T'| = 3/2$,

$$(\mathcal{L}g)(y) = \tfrac{2}{3}\Big[\big(1 + \cos\tfrac{4\pi y}{3}\big)g\big(\tfrac{2y}{3}\big) + \mathbf{1}_{y<1/2}\big(1 + \cos\tfrac{4\pi(y+1)}{3}\big)g\big(\tfrac{2(y+1)}{3}\big)\Big].$$

Power iteration on grids of $2^{10}, \dots, 2^{16}$ gives $1.335037, 1.335065, 1.335071, 1.335073$: six digits of apparent convergence on a number that is not the eigenvalue. The leading eigenvalue is exactly $4/3$, and the argument is one line. The preimages of y are $2y/3$ and $2(y+1)/3$, so $2y/3 = 0$ forces $y = 0$ while $2(y+1)/3 = 0$ forces $y = -1 \notin [0, 1)$; the point 0 receives from nothing but itself, and the adjoint's eigen-equation at an atom there reads $\lambda a = \frac{2}{3}(1 + \cos 0)a = \frac{4}{3}a$. The leading eigenmeasure is atomic — the atom at the fixed point together with the fast-decaying chain on the orbit $(2/3)^n$ into which it leaks — and a grid cannot carry an atom. The discretisation smears it and converges, cleanly and consistently, on the wrong value. Four refinements agreeing to six digits was evidence about resolution, not about the object being resolved. What that atom is has already been identified above: it is the frozen-bit branch of the skew product, and averaging over the bit removes it and leaves a rate of exactly one. The operator here is the one with the bit held at zero, so its $4/3$ is correct and describes a single deterministic branch rather than the octave.

The direct measurement settles it and exhibits the mechanism with no spectral theory in it. Split $\int_0^1 \exp\big(\sum_{m\le K}\varphi(T^m\theta)\big)d\theta$ at $\theta = M(2/3)^K$. On the window, substitute $\theta = (2/3)^K u$; since $\theta(3/2)^m = (2/3)^{K-m}u$, the integrand becomes $2^{K+1}\prod_{j=0}^{K}\cos^2\big(\pi(2/3)^j u\big)$, so the window equals

$$2\Big(\tfrac{4}{3}\Big)^K \int_0^M \prod_{j=0}^{K} \cos^2\big(\pi(2/3)^j u\big)\, du, \tag{9}$$

a K-free constant times $(4/3)^K$. Measured, the window over $(4/3)^K$ reads 0.402071, 0.402035, 0.402029, 0.402028, 0.402028, 0.402028, 0.402028 at $K = 12, 14, \dots, 24$ — flat to six digits across twelve steps, where an eigenvalue of 1.335073 would drift it by 1.6%. Everything outside the window stays bounded over the same range, at $0.32, 0.29, 0.26, 0.23, 0.23, 0.26, 0.27$, and the octave's share of that remainder is flat between 0.015 and 0.028, reproducing from independent code the A_K oscillation measured on the Fourier side.

So the mechanism is elementary. A point with θ of order $(2/3)^K$ keeps its first K iterates near zero, where $1 + \cos$ is 2, and so carries height 2^K; the set of such points has measure $(2/3)^K$; the product is $(4/3)^K$. That is the whole of the growth, and it is an alignment effect at the fixed point rather than anything dynamical. The octave is disjoint from the window for every K past 12, so its flatness is not cancellation — it never had this growth to lose. The 4/3 an earlier draft called the universal obstruction and then withdrew is real and sits here: branching 2 against contraction 2/3, as measure times height at a fixed point, and not as an obstruction to any particular route.

The growth constant is the norm being bounded. The constant in (9) is not a new number. Write $Q(u) = \prod_{j\ge 0} \cos\bigl(\pi(2/3)^j u\bigr)$, so that the window is $2(4/3)^K \int_0^M Q^2$ and $\hat\mu(k) = Q(k/3)$. Since $Q(u) = \hat\nu(u/2)$ for ν the law of $\sum_{j\ge 0} \pm(2/3)^j$, setting $\rho = \nu * \nu$ gives $Q(u)^2 = \hat\rho(u/2)$, whose Fourier transform is $2\rho(2x)$ and is supported in $|x| \le 3$. Poisson summation at spacing $1/3$ then gives, with only $n = 0$ landing inside that support,

$$\|f\|_2^2 = \sum_{k\in\mathbb{Z}} \bigl|\hat\mu(k)\bigr|^2 = \sum_{k\in\mathbb{Z}} Q(k/3)^2 = 3\sum_n \widehat{Q^2}(3n) = 6\sum_n \rho(6n) = 6\rho(0) = 6\int_0^\infty Q(u)^2\,du. \tag{10}$$

Numerically $6\int_0^M Q^2$ reads 1.205739, 1.206198, 1.206295, 1.206307, 1.206310 at $M = 10, 10^2, \dots, 10^5$, against $1+2S = 1.206310$ computed independently on the Fourier side. Hence $\int_0^1 \exp\bigl(\sum_{m\le K} \varphi(T^m\theta)\bigr) \sim \frac{1}{3}\|f\|_2^2\,(4/3)^K$: the constant standing in front of the operator's leading eigenvalue is the very norm whose finiteness is the question. The transfer-operator route is circular in the same way the majorant class is — it returns the quantity it was meant to bound — and this is the same fact as the weight profile recorded below, where $\mathrm{Cum}(u)(4/3)^{-K} \to \frac{2}{3}\|f\|_2^2 u$, reached from the opposite side.

The tail obeys an exact renewal, and L^2 is one decorrelation. Q is self-similar: $Q(u) = \cos(\pi u)\,Q(2u/3)$, so $Q\bigl(\frac{3}{2}v\bigr) = \cos\bigl(\frac{3\pi v}{2}\bigr)Q(v)$ and, substituting $u = \frac{3}{2}v$ in the tail of (10),

$$\int_M^\infty Q(u)^2\,du = \tfrac{3}{2}\int_{2M/3}^\infty \cos^2\bigl(\tfrac{3\pi v}{2}\bigr)Q(v)^2\,dv. \tag{11}$$

This is exact. Were $\cos^2(3\pi v/2)$ to decorrelate from $Q(v)^2$ on tail windows it would contribute its mean $\frac{1}{2}$, the energy would fall by 3/4 at each $\frac{3}{2}$-step, and the exponent would be $\log(4/3)/\log(3/2) = 0.7095$ — the endpoint exponent already recorded here. Membership in L^2 is therefore exactly a statement that this one correlation does not persist, which is the phase question in the only place it can be watched directly across many scales. Measured on the ladder $N_n = (3/2)^n$ up to 10^5 in one sweep, the ratio $\mathrm{corr}_n = \int_{N_n}^{N_{n+1}} \cos^2(3\pi v/2)Q^2 \big/ \frac{1}{2}\int_{N_n}^{N_{n+1}} Q^2$ has mean 0.978 with standard deviation 0.087 over $n = 8, \dots, 27$, a standard error of 0.020, and an ordinary-least-squares slope of $+0.001$ per step: no drift across twenty scales. The

realised block ratios give exponent 0.775, faster than the decorrelated 0.7095 by exactly the amount corr < 1 accounts for, and agreeing with the octave energy $E(R) \approx R^{-0.78}$ measured by unrelated code.

The power of that null matters more than the null. A correlation biased high by d multiplies the per-step ratio by d and costs $\log d/\log(3/2)$ of exponent: $d = 1.05$ costs 0.12 and sits 3.6 standard errors away, while $d = 1.35$ — which would stop the decay outright — sits at twenty. So the flat reading is informative rather than merely quiet. What it is not is a proof: corr_n fluctuates with standard deviation 0.087 per block, twenty scales of a single curve are not twenty independent draws, and the permanent limit stands — over any reachable range, bounded and slowly divergent cannot be told apart. Unwound on blocks the renewal gives a threshold, and that is the sharpest form of the question here. Writing $B_n = \int_{N_n}^{N_{n+1}} Q^2$ and C_n for the same integral against $\cos(3\pi v)$, with $N_n = (3/2)^n$ and $\rho_n = C_n/B_n$, (11) on a single block reads

$$B_n = \tfrac{3}{4}\big(1 + \rho_{n-1}\big)B_{n-1}, \qquad \text{so} \qquad B_n = B_0 \prod_{k<n} \tfrac{3}{4}\big(1 + \rho_k\big), \tag{12}$$

verified against the measured ladder at all nineteen consecutive pairs to $8 \cdot 10^{-5}$, which is the rounding of the tabulated values. Membership in L^2 is the summability of that product, and the decay stops exactly when $\frac{3}{4}(1+\rho) = 1$: the critical value is $\rho = 1/3$.

That $1/3$ is the same number as the half-shift identity, and it is worth being exact about how little that says. Summing $V_n = \frac{4}{3}B_{n+1} - B_n$ over all $n \in \mathbb{Z}$ telescopes, so $\sum V_n / \sum B_n = 1/3$ for *any* positive summable sequence B_n: the whole-line ratio is forced by the renewal alone and carries no information about correlation. It coincides with the critical value for the same reason — $1/3$ is where the multiplier $\frac{3}{4}(1+\rho)$ equals one — so $\rho = 1/3$ is the neutral point of the recursion, pinned at $M = 0$ by an identity that reads $A(0) = A(0)$, and free at every other scale. An earlier draft of this section read the coincidence as the bulk sitting at criticality; that was a tautology dressed as a fact, and what actually matters is only how ρ behaves in the tail. Measured, it falls fast: the tail cross-ratio reads $0.032, -0.039, -0.008, -0.041$ at $M = 10, 10^2, 10^3, 10^4$, and over the blocks $n = 8, \dots, 27$ the mean is -0.0224 with block standard deviation 0.0860 and standard error 0.0192. The margin to the threshold is 0.356, about four block standard deviations. What is not available is any reason it must stay there.

An equivalent reformulation, and what validates it. Write $c(n) = B_n^{-1} \int_{N_n}^{N_{n+1}} e^{3\pi i v} Q(v)^2 dv$, so that $\rho_n = \mathrm{Re}\, c(n)$. Then (12) gives

$$\text{if} \quad \mathrm{Re}\, c(n) \le \tfrac{1}{3} - \delta \ \text{ for all } n \ge n_0, \quad \text{then} \quad B_{n+1} \le \big(1 - \tfrac{3}{4}\delta\big)B_n, \quad \text{so} \quad \sum B_n < \infty,$$

and μ has an L^2 density. It must be said plainly what this is and is not. By (12), $\rho_n = \frac{4}{3}B_{n+1}/B_n - 1$ identically, so the hypothesis is the conclusion in other coordinates: it is an equivalent reformulation, not a reduction, and the statement above reads "if the blocks decay geometrically then the series converges". Three further limits belong with it. It is sufficient only — $B_n = n^{-2}$ is summable with $\rho_n \to 1/3$ and no δ, so a failure of the hypothesis refutes nothing. Only the real part is needed, so bounding $|c(n)|$ instead is strictly harder than the criterion it would establish. And it targets L^2, which is strictly stronger than absolute continuity, so μ could be absolutely continuous with $\sum B_n$ divergent.

What survives is worth having anyway, because the reformulation gives ρ_n a meaning the block ratio does not wear on its face: it is a windowed Fourier coefficient, so "the blocks decay"

becomes "the mass of $Q^2 dv$ on $[(3/2)^n, (3/2)^{n+1})$ is not aligned with the phase $v \bmod \frac{2}{3}$". A windowed equidistribution statement is a legitimate proof target even when it is equivalent to its conclusion, and it is a target of a different shape from anything else in this note.

Measured out to $u = 1.5 \cdot 10^6$ — thirty-four active cosine factors — $|c(n)|$ has mean 0.112 over $n \geq 20$ and mean 0.080 with maximum 0.115 over $n \geq 28$. Reproduced here independently at $n = 28, 30, 32$: 0.0386, 0.1151, 0.0700, with grid spacings $h = 1/16$ and $h = 1/48$ agreeing to five decimals and B_n moving by 10^{-12}, so the quadrature is converged and the discretisation failure that produced the spurious 1.335 above is absent here.

The right way to test a statistic like this is not a standard error but a control at parameters whose answer is known, and it passes. At $\lambda = 1/\varphi$, a Pisot number where μ is singular and $\sum B_n$ must diverge, the same construction over the same range gives mean $\rho = +0.2404$ against its own critical value $+0.2361$, with B_n flat to rising and ρ_n stable to eight digits under grid refinement — the statistic reads criticality to three digits where criticality is the truth. At $\lambda = 2^{-1/2}$, a Garsia number with a bounded density, it gives mean $\rho = -0.4991$ and twenty-six e-folds of decay. At $2/3$ it gives mean $\rho = -0.0231$ with 6.3 e-folds, sitting with the convergent case and 0.356 below its own critical value. So the statistic is not blind, and that control is worth more than any variance estimate computed inside a single deterministic curve. One limitation outranks the others and should close this section, and it is worse than it first appears. Every statistic here has been given standing by the same two controls: $\lambda = 1/\varphi$, where the measure is singular, and $\lambda = 2^{-1/2}$, where it has a bounded density. Both are algebraic, and the singular one is Pisot — and by Erdős's theorem Pisot implies singular, while no non-Pisot singular parameter in $(1/2, 1)$ is known at all. So the controls demonstrate that these statistics detect *Pisot* criticality and nothing else.

Even that overstates them. At a Pisot λ the criterion is forced by an identity rather than detecting anything: the cell at $\xi = 1$ has $\|\lambda^{-m}\| \to 0$ geometrically, so its total cost is bounded while its width is of order λ^K, and that single cell already gives $I_K \geq c\,(2/\lambda)^{-K}$, which is exactly the threshold. The observed near-equality at φ — failing by 0.0037 — is therefore a structural lower bound wearing the costume of a validation, and its flatness is the flatness of one localised cell rather than of an extensive sum. At $2/3$ the same quantity wobbles by 0.028 per step, fifty times larger, because there the sum lives on hundreds of cells. The control tells us nothing about the measurement at issue.

So if μ at $2/3$ is singular, it is singular for a reason with no exhibited example, against which the measured power of every null in this note is not merely unknown but, as the subject stands, unmeasurable. That $2/3$ sits with the absolutely continuous case is a statement that it is not Pisot-like, which was never in doubt — and $2/3$ also sits exactly at the boundary $3\lambda = 2$, where a smoothed escape rate equals λ identically, so at that one parameter a correct computation and a broken one agree.

One register, too small. The obstructions above were found separately and read as a list; they are one arithmetic fact. Everything in the product is a function of ξ, hence of the pair (A_m, x_m) with $A_m = \mathrm{round}((3/2)^m \xi)$. The parity bit is $A_m \bmod 2$, while the depth-k phase history is determined by x_m together with $A_m \bmod 3^k$, since $x_{m-1} = \{((2A_m \bmod 3) + 2x_m)/3\}$ and that iterates. So the entire coupling between bit and history is governed by whether A_m equidistributes modulo $2 \cdot 3^k$ — and on the octave A_m ranges over only about $(3/2)^m$ consecutive integers.

Three consequences follow, and they are the three obstructions. The free window is exactly

$k < m \log(3/2)/\log 3 = 0.36907\, m$, the depth at which 3^k still fits inside the available range: that is the constant measured earlier, now explained rather than observed. The full history wants 3^m residues against $(3/2)^m$ supplied, a shortfall of 2^m; and $\log 3 - \log(3/2) = \log 2$ exactly, which is simultaneously the gap between the entropy rate of orbit-sampled digit statistics and that of the leaf strings the criterion runs over. Against the budget, $\log 2/\log(4/3) = 2.4094$, so the deficit is not marginally too large but larger by a factor of two and a half, in every formulation.

One quantity should not be folded into that. Dropping the parity constraint does not merely admit 2^K times as many strings: the parity-dropped sum with the exact cost is $(3/2)^K$ identically, a rate of 0.40547, so the loss is $0.40547 - \Phi \approx 0.7255$ rather than $\log 2 = 0.69315$. The extra 0.032 is the weight bias of the admitted strings, which carry more than the true ones. The information deficit is exactly $\log 2$; the relaxation penalty is that deficit plus a bias, and calling them one number is tidier than the arithmetic allows.

What this bounds, and what it does not, is worth being exact about. It bounds any argument that *decouples* the bit from the deep history: such an argument asks A_m to equidistribute past its own range and pays $\log 2$ per step, whatever its cost model, cost order or automaton size. It says nothing against an argument that keeps them coupled and estimates the actual correlation, whose measured per-step defect is about 0.048 — far below what any worst-case relaxation must pay. That remaining route is the decoupling statement for a leaf of the 2-adic solenoid, and the deficit explains precisely why it cannot be reached by relaxation, only by genuine equidistribution input about $(3/2)^m$.

Two structural cautions belong beside it, both established here on a second operator. The killed transfer operator for $x \mapsto \lambda x + \eta$ on its trapping window has escape rate exactly λ once $3\lambda \leq 2$, because each branch preimage has length 4λ and fits inside the window; so any smoothed version of the problem returns the critical coincidence by arithmetic, at every resolution rather than in a limit. An exact-overlap discretisation confirms it, giving λ flat to six digits at grids of 500 through 32000 — and a floating-point particle filter converges confidently to 0.618014 ± 0.000039, which is λ to five places, erasing the 0.37% signal it was built to see, while the exact filter recovers 0.6202986. Smoothing does not degrade the answer gradually; it replaces it. That is the same failure as the spurious 1.335073 above, in a different operator, now with an analytic reason behind it. (An earlier draft reported a grid ladder drifting away from the truth; that does not reproduce, and the correct statement is the stronger one just given.)

There is a second reason to read the escape rate at 2/3 narrowly. That parameter sits exactly at the boundary $3\lambda = 2$, where the smoothed rate equals λ identically, so a correct computation and a broken one return the same value there; and $\rho = \lambda$ is the first-moment count already known to be exactly critical, since the cells per unit and the mean cell length multiply to one at every level. A Pisot-sized excess is therefore excluded at 2/3 by a statistic validated on Pisot parameters, at a place where Pisot behaviour was already ruled out by the dissociativity above. It is a consistent reading, not an informative one.

Its limit is exact and should be stated as such. The Pisot control has no transient — ρ_n is essentially constant from the first block — so it establishes that the statistic detects criticality that is present from the start, and says nothing about a measure which mimics decorrelation for thirty scales and then re-correlates. What the measurement excludes is therefore the onset of re-correlation before $n \approx 34$; onset after that is untouched, and no finite ladder of blocks can touch it. An earlier draft of this section reported a drift in the segment means and exponents

as the signature such a tail would leave. That is withdrawn: the segments were nested, one block carries about sixty per cent of the apparent gap, and on disjoint windows there is no gradient at all. There is a structural reason $|c(n)|$ is small, and it is the same arithmetic that runs through the rest of this note. Rescale the block by $v = (3/2)^n w$ with $w \in [1, 3/2)$. Since $Q\big((3/2)^n w\big) = \prod_{i=1}^{n} \cos\big(\pi (3/2)^i w\big)\, Q(w)$, the integrand becomes a Riesz product of ratio $3/2$,

$$\prod_{i=1}^{n} \cos^2\big(\pi (3/2)^i w\big) = \prod_{i=1}^{n} \tfrac{1}{2}\big(1 + \cos(2\pi (3/2)^i w)\big),$$

weighted by $Q(w)^2$, and $c(n)$ is its normalised Fourier coefficient at frequency $(3/2)^{n+1}$. That frequency is not in the spectrum. The coefficients live on $\sum_{i \le n} \varepsilon_i (3/2)^i$ with $\varepsilon_i \in \{0, \pm 1\}$; were $(3/2)^{n+1}$ such a sum, clearing denominators would give $3^{n+1} = \sum_{i \le n} \varepsilon_i 3^i 2^{\,n+1-i}$, and reducing modulo 3 kills every term but $i = 0$, forcing $\varepsilon_0 = 0$; dividing by 3 reproduces the same equation one degree lower, so every ε_i vanishes and $3^{n+1} = 0$. An exhaustive search over $\{0, \pm 1\}^{n+1}$ confirms it for $n \le 9$.

So $c(n)$ would be exactly zero on a full period, and what the block measures is only the leakage from the finite window $[1, 3/2)$ and the weight $Q(w)^2$. That is why the measured values sit near 0.08 rather than near 1. It is an explanation and not a bound, and the gap between the two is the one this note keeps arriving at: expanding the product into its 3^{n+1} characters and estimating the window transform against each overshoots by exactly 2^n, since the coefficients have total variation $\prod(1 + \frac{1}{2} + \frac{1}{2}) = 2^{n+1}$ against a mean of 1. Nor does exactness help by itself — the distance from $(3/2)^{n+1}$ to the nearest attained sum is $|3^{n+1} - \sum \varepsilon_i 3^i 2^{\,n+1-i}|/2^{\,n+1}$, whose numerator is a nonzero integer and so at least one, which bounds the distance below only by $2^{-(n+1)}$. Lines may therefore approach the target exponentially closely, and a proof must count how many do, not merely observe that none is exact. That is the same distinction already drawn for the dissociateness lemma above. That count can be done exactly, and it settles the matter against this line of attack. Writing $S_\varepsilon = \sum_{i \le n} \varepsilon_i 3^i 2^{\,n+1-i}$, the number of $\varepsilon \in \{0, \pm 1\}^{n+1}$ with $|3^{n+1} - S_\varepsilon| \le D\, 2^{\,n+1}$ — that is, with frequency distance at most D — is, by meet-in-the-middle at every n from 4 to 16 and every $D \in \{\frac{1}{2}, 1, 2, 4\}$, exactly

$$N(n, D) = D\, 2^{\,n},$$

with the smallest attained integer gap equal to 1 throughout, so the nearest line lies at frequency distance $2^{-(n+1)}$. And $D2^n$ is precisely what uniform spreading predicts: 3^{n+1} values spread over a range of about $4 \cdot 3^{n+1}$ have density $\frac{1}{4}$ per integer, giving $2D2^{\,n+1}/4 = D2^{\,n}$. The measured count matches to the last digit at every n.

So there is no arithmetic conspiracy near this frequency and no Diophantine obstruction to be found; the sum-set is as evenly spread as a random set of the same size, and the difficulty is density rather than arithmetic. The consequence is negative and worth stating plainly: counting cannot prove the criterion. Any estimate that treats the leakage line by line in absolute value sums 2^n terms of size $O(1)$ and fails by exactly the factor that closes every linearising route in this note — the same 2^n, reached now from a third direction. What the criterion needs is cancellation among the leakage phases. That leaves it a genuine reformulation of the question rather than a way around it, which is the honest description of everything in this section.

Erdős–Kahane at a fixed parameter: the count is an identity, but the route is live. The one remaining route that does not linearise survives, but not in the form it is usually given: its counting step is empty here, and what carries any decay is the per-cell charge instead. The apparatus is exact arithmetic here. With $x_n = (3/2)^{n-N}\xi$, $A_n = \mathrm{round}(x_n)$ and $d_n = x_n - A_n$,

$$e_n := 2A_{n+1} - 3A_n = 3d_n - 2d_{n+1} \in \{0, \pm 1, \pm 2\}, \qquad e_n \equiv A_n \bmod 2, \tag{13}$$

checked in rational arithmetic over 12,000 steps without a failure, so the digit string of ξ is the itinerary of a full-branch expanding map of slope $3/2$ and ξ is its initial condition.

The engine of the method is that the number of admissible strings times the parameter measure carried by each tends to zero. Over λ that measure is exponentially small. At a fixed λ it is not: the number of cells per unit of ξ is $\sum_m (2/3)^m = 3$ exactly, independent of N, the mean cell length is $1/3$, and cylinder measure times contraction is $(3/2)^N(2/3)^N = 1$. Three independent derivations agree, so the product is exactly 1 and the counting step does no work at any N. The mechanism is that ξ enters its own orbit through a single linear coordinate, $dx_n/d\xi = (3/2)^{n-N} \le 1$, so the level-$n$ constraints are nested and intersect to the tightest one; over λ they are transversal and multiply. Rationality of $1/\lambda = 3/2$ is what supplies the exact finite alphabet in (13) and simultaneously empties the count.

Two numerical facts are worth keeping. Membership in L^2 needs a cost exceeding $\frac{1}{2}\log(3/2) = 0.20273$ nats per step; the support-only version of the method delivers 0.1832, and a version aware of $|e_n|$ delivers a mean of 0.304 — half again what is required. And yet the bound it actually yields, $R\,\mathbb{E}[e^{-2\,\mathrm{cost}}]$, *grows*, as $R^{+0.126\pm0.027}$; the support-only version grows as $R^{+0.34}$; the truth decays as $R^{-0.8}$. Using $e^{-2\mathbb{E}[\cdot]}$ in place of $\mathbb{E}[e^{-2\cdot}]$ is Jensen's inequality in the wrong direction, and here it reverses the sign of the conclusion rather than merely weakening it: the annealed average is dominated by the rare strings that avoid $|e_n| = 2$ entirely. It is worth recording as a general caution, since the mean cost comfortably clears the target while the method fails.

What that does *not* establish is that the route is closed, and an earlier draft of this paragraph said it was. The claim was that any cost function good enough to decay must encode the cylinder geometry and would therefore be evaluating the integral rather than counting strings. Exact cell enumeration refutes it. Charging each maximal run of nonzero e_n its own constrained optimum — a relaxation, since dropping the couplings between runs only raises the maximum and so lowers the charge, giving a valid majorant, audited directly at $+1.1\cdot10^{-13}$ over all 82,738 runs at $N = 21$ — produces a bound that *decays*, at $R^{-0.28}$ to $R^{-0.31}$, falling monotonically at every N from 11 to 30 and by a factor of 9.1 over nineteen octaves. On the identical machinery the per-symbol cost grows at $R^{+0.11}$, so the test could plainly have come out the other way; and the exact enumeration removes the sampling caveats that attended the earlier Monte Carlo estimate of the same growth.

The crossover sits at order two. Charging only disjoint blocks of at most two consecutive jump symbols already decays, at $R^{-0.15}$, and order six is within two per cent of the full cost. That cost is a strictly local, finite-order, purely combinatorial function of the digit string: it discards every coupling between runs and everything at the roughly one third of indices carrying $e_n = 0$, and at $N = 25$ it sits some $3\cdot10^5$ above the true octave energy while decaying at less than half the true rate. It is not the integral in disguise.

So the route is live, and the obstruction is not where it appeared to be. It is not the counting step — that really is an identity, and no variant will draw decay from it — but the free energy of the digit process. What a theorem needs here is a rigorous large-deviation estimate

for the run-pattern statistics of the actual $\times 3/2$ orbit, and that process is demonstrably not independent: an independent model misses the measured per-symbol exponent by about 0.05 nats per step, in the direction and of the size that the measured over-dispersion of the $|e_n| = 2$ counts predicts. A numerically decaying bound over nineteen octaves is a measurement, not a proof, and these are nested octaves rather than independent observations, so what carries the sign is the strict monotonicity rather than any standard error.

The recursion closes on the integers, and that is why it cannot help. Write $T(f) = \int_{\mathbb{R}} \cos(2\pi f v) Q(v)^2 dv$. The self-similarity $Q(v)^2 = \cos^2(\pi v) Q(2v/3)^2$ together with $\cos^2(\pi v) = \frac{1}{2} + \frac{1}{2}\cos 2\pi v$ gives an exact three-term relation,

$$T(f) = \tfrac{3}{4} T\big(\tfrac{3f}{2}\big) + \tfrac{3}{8} T\big(\tfrac{3(f+1)}{2}\big) + \tfrac{3}{8} T\big(\tfrac{3(f-1)}{2}\big). \tag{14}$$

Since $T(f) = 2\rho(2f)$ and $\rho = \nu * \nu$ is supported in $|x| \le 6$, T vanishes for $|f| > 3$; the three-term step at an integer reaches only $0, \frac{3}{2}, 3, \frac{9}{2}$, and that truncation closes the system. Taking $f = 0, 1, 2, 3$ in turn gives $T(3/2) = T(0)/3$, then $T(1) = \frac{5}{8}T(0)$, then $T(2) = \frac{1}{8}T(0)$, with $T(3) = 0$ consistent. Measured on the direct integral to $M = 10^5$: 0.62500006, 0.33333349, 0.12500006, and $T(3), T(7/2), T(4)$ zero to 10^{-15}. The points that do not close look nothing like rationals, $T(1/2)/T(0) = 0.88359189$ and $T(5/2)/T(0) = 0.02328172$, which is what the escape into quarters predicts.

In terms of the autocorrelation itself, with $A(s) = 3T(3s)$, these read $A(1/3) = \frac{5}{8}A(0)$, $A(1/2) = \frac{1}{3}A(0)$, $A(2/3) = \frac{1}{8}A(0)$ and $A(1) = 0$. The middle one is the half-shift identity; the outer two came from the same recursion. The last is free, since f lives on $[0, 1]$, and so serves as a check on the correspondence rather than as a result. A second check needs no computation at all: A is even, supported in $[-1, 1]$, and integrates to $(\int f)^2 = 1$, so $\int_0^1 A = \frac{1}{2}$ exactly, and the four predicted values $1.206310, 0.753944, 0.150789, 0$ give 0.502629 under the trapezoid rule at $h = 1/3$ — inside its own error on four points, and overshooting as it must for a convex A. All four are confirmed at once by a constraint none of them was derived from.

None of it can settle the question, and the reason is structural rather than technical. Equation (14) is linear and homogeneous in T, so it determines T up to a single scalar, and that scalar is $T(0) = \|f\|_2^2/3$ — the unknown. Every identity in this family constrains the shape of the autocorrelation and none constrains its scale; and each presupposes what it would prove, since were $\|f\|_2^2$ infinite the relations would read $\infty = \infty$. The measure's defining property is self-similarity, self-similarity generates homogeneous linear relations, and finiteness of the L^2 norm is exactly the inhomogeneous information that self-similarity cannot supply. That is the same fact as the observation running through this note, that every closed route ends in a phase count: the phases are what the functional equation does not carry. A proof must inject something that is not a linear identity — the arithmetic of $(3/2)^m$, or a counting argument on the nonnegative product.

A numerical caution belongs with (8), since it cost an hour. Generating the orbit by iterating $\theta \mapsto \frac{3}{2}\theta \bmod 1$ in floating point multiplies the error by $3/2$ at every step and destroys the fractional part: done that way the right-hand side reads $0.045, 0.097, 0.186, 0.341, 0.614, 1.100$ at $K = 6, \ldots, 16$, growing, against a true value near 0.05. Computing each $\theta_m = \theta(3/2)^m \bmod 1$ in closed form instead reproduces the direct value exactly. A chaotic map must not be iterated numerically when its orbit is available in closed form.

What the peak count would have to supply. Being outside the shape is not the same as being easy, and the requirement can be written down. Cover $[1,2]$, bound P_K on each piece, and sum height against measure. A point meeting a fraction α of the K conditions $\|t\lambda_m\| < \delta$ carries height about $2^{\alpha K}$; if those conditions behaved independently, with $p_0 = 2\delta$, the measure of the set meeting at least αK of them would be $e^{-KD(\alpha\|p_0)}$, and the level would contribute $\big(2^{\alpha}e^{-D(\alpha\|p_0)}\big)^K$. Optimising over α gives the log moment generating function exactly,

$$\sup_{\alpha}\big[\alpha\log 2 - D(\alpha\|p_0)\big] = \log(1+p_0),$$

so the independence estimate is $(1+2\delta)^K$ — which is just the first moment $\mathbb{E}\,2^{\#\mathrm{met}}$, and exceeds one for every threshold. Numerically 1.20^K, 1.30^K, 1.40^K at $\delta = 0.10, 0.15, 0.20$. No choice of δ rescues it.

Against that, the measured carrier has measure 0.7119^K and height 1.4046^K, whose product is 1 by construction, being the flatness of I_K rather than an independent fact. The gap between 0.7119 and the independence values 0.80, 0.9165, 0.9798 at $\alpha = \frac{1}{2}$ is what a proof would have to supply: the conditions $\|t\lambda_m\| < \delta$ must be shown *positively correlated at a definite rate*, not merely counted. That is the honest price of the route that survives, and it is why it runs into the distribution of $(3/2)^m$ rather than around it.

A caution applies to all of these increment statistics, and to several figures earlier in this note. There is a transient below $K \approx 25$, and the range a laptop reaches in reasonable time lies wholly inside it. Measured here, the log-drift of I_K per step is -0.0165 over $K = 7,\dots,26$ and -0.0247 over $K = 15,\dots,26$, against $+0.00009$ over $K = 25,\dots,37$ from the longer run: two regimes, not a disagreement. The same split explains the standard deviation of the relative increments, 0.089 to 0.137 below $K = 26$ against 0.071 above $K = 20$. It is also the common cause of the trend readings withdrawn above — the stride fit, the sign-turning window fits, the declining correlation ratio, and the dispute over whether the correlation accumulates. None of those was a property of the object. Any statement here fitted below $K = 25$ describes the transient.

The mass has a fixed carrier. At $K = 20, 24, 28$ the same peak locations recur — 1.869287, 1.878192, 1.871758, 1.881893, 1.622273 — with heights growing like 1.40^K and widths like $(2/3)^K$, and the set is closed under $t \mapsto 2t/3$, the ratio-3/2 dynamics acting on it. The top 10^{-3} of Lebesgue measure carries 55% of the integral at $K = 20$, 80% at $K = 26$ and 95% at $K = 32$. So $A_K = O(1)$ would follow from counting those peaks and bounding height against width, which is a concrete programme rather than a hope. These measurements are an agent's, on its own code, and their adversarial check had not returned when this was written.

They do cohere, and the check is not circular, because one of the four numbers is forced rather than measured. Height 1.4046^K against width $(2/3)^K$ gives 0.9364^K per peak, so a flat integral requires the number of peaks to grow like 1.0679^K; that count was never measured. It then predicts a carrying set of measure $(2/3\cdot 1.0679)^K = 0.7119^K$, namely $1.1\cdot 10^{-3}$ at $K = 20$, $1.5\cdot 10^{-4}$ at $K = 26$ and $1.9\cdot 10^{-5}$ at $K = 32$ — against the reported concentration, where the top 10^{-3} of Lebesgue measure carries 55%, 80% and 95% at those same three values, and the top 10^{-5} carries 31% at $K = 32$. The predicted carrier crosses 10^{-3} exactly where the reported share stops being partial, and at $K = 32$ sits at about twice 10^{-5}, which is what a 31% share of it wants. Four independently produced quantities, one of them derived from the other three, agreeing at three depths.

It is worth seeing what a peak is, because it says where the counting programme leads. A factor is near its maximum when $t(3/2)^m/3$ is near an integer, so a tall peak is a t at which

many of those quantities are simultaneously small. At the reported top peak $t = 1.869287$, fourteen of the first twenty-eight are within 0.15 of an integer, against eight at the generic point $t = \frac{3}{2}$; and $1.4046^K = 2^{0.4903K}$ is exactly what half the factors contributing a two apiece would give. The threshold is arbitrary and the reading is indicative rather than established — a second listed peak has only five of twenty-eight within 0.15 and is still tall — but the shape is clear enough. Counting the peaks means counting the $t \in [1, 2]$ for which $\|t(3/2)^m/3\|$ is small for a positive proportion of $m \le K$, and that is a question about the orbit of $(3/2)^m$ modulo one: Mahler's problem, where even density is unknown. The programme is concrete, and it is concrete about something hard.

One consequence deserves emphasis because it retires a control used throughout this note. Writing the factors as $1 + c\cos$, the level and volatility of A_K move smoothly through $c = 1$. At $c = 0.6, 0.8, 0.9, 0.95, 0.98, 1$ the means are 0.498, 0.250, 0.140, 0.092, 0.067, 0.052 and the increment standard deviations 0.0035, 0.017, 0.036, 0.055, 0.073, 0.090. The fair coin is not a phase transition for this quantity, so a proof uniform in c would be correct, and the biased-coin substitution — decisive elsewhere here — carries no information about A_K at all.

The limitation is worth stating in full, because it is permanent. Boundedness and logarithmic growth cannot be told apart here by computation. Over $K = 6$ to 40 the quantity $\log K$ spans a factor 2.06, and the observed oscillation spans 2.12: the data has exactly the amplitude a logarithm would have. Separating the two would need K of order 1600, so R of order 10^{280}. Every numerical statement in this note about A_K staying bounded is therefore consistent with $A_K \sim c \log K$, which would sink the route, and no amount of further computing will change that. The freedom to move the window is real but not needed: splitting the octave into m blocks and letting each window slide within its block, the worst block's best constant is 0.09 at $m = 4$ and 0.21 at $m = 16$, both flat in K across $K = 10$ to 30, so a cover with a bounded constant exists — but the direct position is already better than any of them.

No window can remove that growth. Since only an upper bound on $E(R)$ is needed and $F_K \ge 0$, the sharp octave indicator may be replaced by any nonnegative majorant, and a smooth one has a kernel decaying like $|u|^{-p}$ instead of $|u|^{-1}$. It is natural to hope the geometric growth is an artifact of the sharp window. It is not, and the reason is visible without choosing a window. Let $\mathrm{Cum}(u) = \sum_{0<m\le u2^K} c(m)$ be the weight the kernel meets inside $|u|$. Exhaustive enumeration gives $\mathrm{Cum}(u)(4/3)^{-K}$ equal to $0.15156, 0.17050, 0.18577, 0.19095, 0.19514$ at $u = \frac{1}{4}$ and $0.74250, 0.77842, 0.78679, 0.79493, 0.79931$ at $u = 1$, for $K = 6, 8, 10, 12, 14$: a fixed profile, with the successive ratios converging to $(4/3)^2 = 1.7778$ at every u. The profile is moreover linear in u, so the weight density is flat across the block at rate $(4/3)^K$ per unit of u, and the weight already grows like $(4/3)^K$ inside $|m| \le \frac{1}{4}2^K$ alone. Consequently $\sum_m c(m)|G(m/2^K)| \asymp (4/3)^K \int_0^2 |G|$ for any kernel at all: fast decay does not help, and neither does vanishing to high order at the origin, since the mass is not concentrated there. The only escape would be a kernel whose L^1 norm on the block vanishes, and such a function does not majorise the indicator.

What that closes, and what it does not, needs stating carefully, since the two are easily conflated. The absolute sum bounds $\sup_{k_0} |\mathrm{Err}|$, and that supremum genuinely does grow like $(4/3)^K$ — so no term-by-term absolute bound on a nonnegative majorant can work, at any $\lambda \in (1/2, 1)$. But the lane needs one window, $k_0 = R$, and there the quantity is flat: computing Err/L directly from $\sum_{k\in W} F_K(k)$, with no character expansion at all, it stays between -0.916 and -0.959 at every even K from 6 to 30 — -0.936 at $K = 6$, -0.916 at $K = 12$, -0.955 at

$K = 20$, -0.957 at $K = 30$, the last on a window of 191 751 terms. The $(4/3)^K$ enters at the triangle inequality and is not in the lane. So what is closed is every argument that discards the phases; an argument that keeps them and uses the octave position is not touched by any of this.

That closure reads a linear profile as the signature of an L^2 density, so it is worth checking that the reading discriminates rather than always returning the same answer. Run the same estimator on a measure known to be singular: the middle-thirds Cantor measure, of correlation dimension $\log 2/\log 3 = 0.6309$. At radii $0.05, 0.02, 0.01, 0.005, 0.002$ the ratio $\mathbb{P}(|D| \le r)/r$ is $2.398, 2.409, 2.411, 2.413, 2.414$ for μ, flat to 0.7% and converging to $2\|f\|_2^2 = 2.4126$, and $2.86, 3.87, 5.54, 6.64, 9.30$ for the Cantor measure, growing without settling, with local exponent $0.649, 0.654, 0.628, 0.643, 0.641$ against the predicted 0.6309. The estimator separates them cleanly, so the linear profile is a real signature and not an artifact. The ratio is also much the sharper of the two tests at these radii: for μ the naive exponent $\log \mathbb{P}/\log r$ reads $0.708, 0.775, 0.809, 0.834, 0.858$, still well short of 1, because $\mathbb{P} \sim cr$ with $c \neq 1$ makes that estimator converge only logarithmically. A reader who tested only the exponent would conclude the correlation integral is sublinear, and be wrong.

A sharper control is available inside the same family, and it is the one to use. Let $Y_p = \frac{1}{3}\sum_k B_k (2/3)^k$ with $\mathbb{P}(B = 1) = p$. Then $\hat{\mu}_p(3\xi)/\hat{\mu}_p(2\xi) = (1-p) + pe^{-2\pi i\xi}$, which equals 1 at every integer, so (2) holds for *every* p, while for $p < 0.14028$ the measure is singular, its dimension $H(p)/\log(3/2)$ falling below 1. The functional equation therefore carries no information about absolute continuity by itself, and any argument that uses only (2) is refuted by substituting $p = 0.1$. That substitution is the right test for everything in this note, because it changes nothing about the ratio, the lattice, or the period — only the bias. Applying it to the correlation-integral reading, at depth $K = 12$ the ratio $\mathbb{P}(|Y_p - Y_p'| \le r)/r$ across $r = 0.05, \ldots, 0.002$ is

$p = 0.5$	$2.398, 2.410, 2.414, 2.413, 2.404$	flat
$p = 0.2113$	$3.81, 4.39, 4.83, 5.19, 5.64$	growing
$p = 0.10$	$7.82, 12.77, 18.19, 25.36, 41.15$	growing sharply

so the reading separates $p = 1/2$ from the singular case at the same ratio and the same lattice. It also carries a warning for the second moment below: $p = 0.2113$ is exactly the auxiliary bias that appears there, and its correlation integral is visibly not linear at this depth.

The profile also names itself, which says what the growth really is. Since $c(m) = 2^{K+1}\mathbb{P}(V = m)$ below, $\mathrm{Cum}(u) = 2^K\,\mathbb{P}\big(0 < |3(Y - Y')| \le u(2/3)^K\big)$, so a linear correlation integral forces $\mathrm{Cum}(u) \sim \frac{2}{3}\|f\|_2^2\, u\,(4/3)^K$. Measured at $u = 1$ the sequence extrapolates geometrically to 0.804446 against $\frac{2}{3}\|f\|_2^2 = 0.804207$. The identity $c(m) = 2^{K+1}\mathbb{P}(V = m)$ itself holds exactly, to 0 in floating point over all 3^{K+1} patterns at $K = 3, 5, 7$. So the $(4/3)^K$ is not an obstruction sitting between us and the theorem: it is the theorem, seen in the size of the absolute sum, and a proof must cancel exactly that much and no more.

The circle closes exactly here, and it is worth seeing how. The weight is a probability: $c(m) = 2^{K+1}\mathbb{P}(V = m)$ where $V = \sum_i \eta_i 2^i 3^{K-i}$ with η_i independent, taking 0 with probability $\frac{1}{2}$ and ± 1 with probability $\frac{1}{4}$. But $\eta = B - B'$ for independent fair bits, so

$$V/3^K = \sum_i \eta_i (2/3)^i = 3(Y - Y'), \qquad Y, Y' \text{ independent copies of } \mu.$$

Checked against the exact law at $K = 9$ and a Monte Carlo for $3|Y - Y'|$: the two agree to three digits at every radius from 0.5 down to 0.02. So the local limit theorem the reformulation asks

for is a statement about $\mathbb{P}(|Y-Y'|\le r)$, the correlation integral of μ, and the bound it needs is $O(r)$ exactly, which is $\mu\in L^2$. Measuring that integral from the exact law at $K=11$, the ratio $\mathbb{P}(|3(Y-Y')|\le r)/r$ is 0.7702, 0.7969, 0.8021, 0.8038, 0.8042 at $r=0.5$, 0.2, 0.1, 0.05, 0.02, flat until the truncation scale $(2/3)^{11}=0.0116$ is reached. For an L^2 density the slope should be $\frac{2}{3}\|f\|_2^2=\frac{2}{3}\cdot 1.206310=0.80421$, which is the measured value to four digits: the identification, the L^2 norm and the correlation integral all agree. A finer scan, 400 radii spaced logarithmically at $K=12$, gives $\mathbb{P}(|Y-Y'|\le r)/r$ with maximum 2.4154 against the predicted $2\|f\|_2^2=2.4126$, and over $r<0.1$ the whole range of the ratio is 2.3702 to 2.4154, an oscillation under two per cent. Repeating at $K=14$, which reaches radii a factor of two finer, gives the same window 2.3702 to 2.4139. There is no sparse sequence of radii on which the ratio grows; whatever a counterexample would look like, it does not show here. The reformulation is therefore genuinely a change of coordinates and nothing more; but it is a change of coordinates in which the cancellation has a visible cause rather than being a coincidence among 2^{K+1} phases.

The correlation integral has its own exact recursion, which is worth recording because it says where the linear behaviour comes from. Since $3(Y-Y')=\eta+2(Y-Y')'$ in law, with $\eta=B-B'$ independent of the copy, writing $C(t)=\mathbb{P}(|Y-Y'|\le t)$ and $M(t)=\mathbb{P}(\,|\,|Y-Y'|-\frac{1}{2}|\le t)$ gives

$$C(u)=2\,C(2u/3)-\tfrac{1}{2}M(u),$$

which holds against the exact law at $K=11$ to one part in 10^4 at every radius from 0.4 down to 0.03. Read downward it is a contraction. Writing $\kappa(u)=C(u)/u$ and $m(u)=M(u)/2u$, the same relation is

$$\kappa(2u/3)=\tfrac{3}{4}\big(\kappa(u)+m(u)\big),$$

verified to 10^{-4} at every radius from 0.4 to 0.02. So passing to smaller scales contracts by $3/4$ with m as source, and κ stays bounded if and only if m does; the fixed point is $\kappa=3m$, and measured $\kappa\to 2.4126$ against $m\to 0.8042$, which is that relation to five digits. That fixed point is worth a second look, because it is the same statement as (4) reached from a different direction, and it is what pins the factor $\frac{1}{2}$ in the recursion. If A is the autocorrelation density then $C(u)\sim 2A(0)u$ and $M(u)\sim 4A(\frac{1}{2})u$, so the linear limit of the recursion reads $2A(0)=\frac{4}{3}\cdot 2A(0)-\frac{1}{2}\cdot 4A(\frac{1}{2})$, that is $A(0)=3A(\frac{1}{2})$, which is (4) exactly. Clearing denominators the law at depth K lives on the lattice $3^{-(K+1)}$ and can be built by integer shift-adds; at $K=13$ this gives $C(u)/M(u)=1.500107$ at $u=0.005$, with $C(u)/u\to 2.4132$ and $M(u)/u\to 1.6087$ against the predicted $2\|f\|_2^2=2.412620$ and $\frac{4}{3}\|f\|_2^2=1.608413$. The recursion written with M in place of $\frac{1}{2}M$ would force $C/M\to 3$ and $A(0)=6A(\frac{1}{2})$, so the identity excludes it by a factor of two. The linear behaviour of C is therefore driven entirely by the mass near the half-integer translate, and the question has moved rather than closed. The source itself satisfies a recursion involving further translates, and one can say exactly how badly the resulting system fails to close. Writing $C_a(u)=\mathbb{P}(|Y-Y'-a|\le u)$, the same computation gives

$$C_a(u)=\sum_{\eta}p_\eta\,C_{(3a-\eta)/2}(3u/2),$$

so the translate moves by $a\mapsto(3a-\eta)/2$, which *expands* by $3/2$. Starting from $a=0$ the reachable translates inside $[-1,1]$ number $3,7,15,31,\dots,2^{n+1}-1$ after n steps, and they are exactly the dyadic rationals $j/2^n$: the orbit generates a dense set. The system over translates is therefore infinite-dimensional and dense, not finite, which is the structural reason the transfer-operator picture over translates does not close. Controlling C at the origin requires controlling μ near every dyadic point at once.

The system has a name once one passes to densities. Writing $A(a) = \int f(x)f(x+a)\,dx$ for the autocorrelation, $C_a(u) \sim 2uA(a)$ as $u \to 0$, and the translate recursion becomes a functional equation for A alone:

$$A(a) = \tfrac{3}{2} \sum_{\eta} p_\eta \, A\Big(\frac{3a-\eta}{2}\Big). \tag{15}$$

Measured at $u = 0.02$ and $K = 13$ it holds to $7{\cdot}10^{-4}$ at every a tried, and $A(0) = 1.2043$ against $\|f\|_2^2 = 1.206310$. At $a = 0$ the two outer terms coincide by symmetry and (15) collapses to $A(0) = 3A(\frac{1}{2})$, which is exactly the half-shift identity (4): measured $A(0)/A(\frac{1}{2}) = 2.9930$. So (4), the family of relations after it, and the whole translate system are one equation read at different points, and the question is whether (15) admits a bounded solution, which is $f \in L^2$.

Discretising the operator behind (15) on grids of 301 and 601 points gives a top eigenvalue of 0.99996 with the next moduli near 0.76 and 0.80: eigenvalue one with a gap of about 0.2, which is why the numerics converge so cleanly. It settles nothing, and the reason is worth stating, because it is the same trap as assuming a density: a finite grid imposes boundedness by construction, and the discretised operator is a nonnegative matrix whose Perron eigenvector exists whatever the continuum does. What the computation does say is that if a solution exists in a space where the discretisation converges, it is unique and stable.

The eigenvalue is exactly one, not approximately. Applying the operator to the constant and integrating, the three branches contribute $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ after accounting for the mass that leaves $[-1, 1]$, and the total is exactly 1; the 0.99996 above is grid error. So the constant functional is a left eigenvector at eigenvalue one, and 1 lies in the spectrum unconditionally. This sharpens the statement of the problem as far as it will go: the autocorrelation *measure* of μ always satisfies (15), whatever μ is. What is open is only whether that measure has a bounded density.

Iterating (15) on a grid of 4001 points from the constant function converges in 200 steps to a residual of $2 \cdot 10^{-9}$, and the normalised limit is $A(0) = 1.2063$, $A(\frac{1}{4}) = 0.9225$, $A(\frac{1}{2}) = 0.4021$, $A(\frac{3}{4}) = 0.0749$, against the values $1.204, 0.922, 0.402, 0.075$ measured from the law directly. In particular it returns $\|f\|_2^2 = 1.2063$, agreeing with the meet-in-the-middle value 1.206310 to five digits: a fifth independent route to that number. The same caveat applies as above, that the grid supplies the boundedness the continuum question is about.

That caveat can be tested, and it is the sharpest numerical evidence in this note. If the continuum fixed point were a singular measure, refining the grid would make its peak grow: the value at the origin would behave like N^{1-d} for a local dimension $d < 1$. Running the iteration at $N = 501, 1001, 2001, 4001, 8001, 16001$, a refinement by a factor of 32, gives $A(0)/\int A$ equal to 1.206361, 1.206323, 1.206316, 1.206312, 1.206311, 1.206311, and the supremum equals the value at the origin throughout. It converges, to the meet-in-the-middle value 1.206310 to six digits, rather than growing. This is not a proof, since the interpolation smooths at scale $1/N$ and a sufficiently slowly growing singularity would survive such a test, but it is the test one would run to look for one, and nothing appears.

Counting the branching explains the constant that appears everywhere in this note. Of the three children of each translate, two stay inside $[-1, 1]$ and one escapes, so the active set doubles at every step while the scale contracts by $2/3$; the product is $2 \cdot \frac{2}{3} = \frac{4}{3}$. That is the same $4/3$ by which the size-only bound overshoots, the same $4/3$ separating $\frac{1}{2}$ from $\frac{2}{3}$ in the Riesz exponent, and the same $4/3$ in the allowance a window bound would be given. The routes are not failing for unrelated reasons: they are all reading one branching-times-contraction ratio that sits on the wrong side of one.

A test function returns the autocorrelation. The cleanest statement of why the majorant idea cannot work needs no truncation, no window shape and no counting. Since the $|\hat{\mu}(k)|^2$ are exactly the Fourier coefficients of the autocorrelation density A, for any $M : \mathbb{Z} \to \mathbb{R}$ with enough decay,

$$\sum_k M(k)|\hat{\mu}(k)|^2 = \int_0^1 A(t)\,\check{M}(t)\,dt, \qquad \check{M}(t) = \sum_k M(k)e(kt). \tag{16}$$

So every nonnegative test function dominating 1_W returns an integral of A against its own transform: it does not bound the correlation integral, it hands it back. Checked at $R = 4096$ with an explicit raised-cosine majorant, the two sides agree to a relative $2 \cdot 10^{-12}$, and the series for A returns $A(0) = 1.206310$, a third independent route to $\|f\|_2^2$.

The earlier version of this paragraph set the collapse threshold at $Q \asymp (3/2)^K$ and observed that $\min_{\varepsilon \neq 0} |\nu_\varepsilon| = 1$ for every $K \leq 22$, flat, while $(3/2)^K$ runs to 7482; that verdict was right but its threshold was too generous, since efficiency forces $Q = dP \geq P/L = 3 \cdot 2^K$, larger by $3(4/3)^K$. Dissociateness says no combination vanishes; it says nothing about how near zero one can come. One caution on the other side, since it was proposed and does not hold: there is no per-octave floor of $2\|f\|_2^2$ for majorant bounds. That would follow from $\int_{-d}^{d} \check{M} = M(0) \geq 1$, but an octave window does not contain the origin, so domination gives no lower bound at $k = 0$ — measured, $M(0) = 0$ for the majorant above, which nonetheless costs only a factor 1.59 over $E(R)$ itself. Majorants are not lossy here; they are simply circular.

A size-only bound cannot work. Bounding Err by absolute values exceeds the requirement by a factor growing like $(4/3)^K$, matching $\sum_\varepsilon c_\varepsilon = 2^{K+1}$ against a requirement of order $(3/2)^K$.

How much cancellation there is. Computing the signed Err exactly, two ways that share no code and agree to six digits:

K	3	5	7	9	11	12	13
Err$/L$	-0.867	-0.978	-0.949	-0.936	-0.925	-0.917	-0.938
bound$/L$	10.8	26.0	63.4	140.2	295.5	422.7	606

The quantity a proof must bound above is flat at about -0.93, and negative, so the window sum falls below its main term. Over the eleven values its mean is -0.926 with standard deviation 0.038, and a linear fit in K has slope -0.0046 against a standard error of 0.0036, so the trend is indistinguishable from flat; the window-to-period ratio averages 0.074. The only available bound on it is off by a factor 646 by $K = 13$ and grows geometrically. Both methods are limited by the period and stop at $K = 13$.

The octave form is an equivalence, not a reduction. (The symbol here is local: $\widetilde{I}_K$ denotes a different normalisation from the octave integral I_K of the main text.) With $\widetilde{I}_K = \int_1^{3/2} P_K^2$, the bound $\widetilde{I}_K \leq Cq^K$ with $q < 2/3$ implies $\mu \in L^2$, and $\mu \in L^2$ implies $\widetilde{I}_K \leq C(2/3)^K$ by a Fejér minorant. The substitution $y = (3/2)^K x$ makes this visible in one line:

$$\widetilde{I}_K = (2/3)^K \int_{(3/2)^K}^{(3/2)^{K+1}} \prod_{i=0}^{K} \cos^2(\pi(2/3)^i y)\,dy,$$

so $\widetilde{I}_K$ is $(2/3)^K$ times a Fourier energy over one octave. Checked to six figures at $K = 6, 10, 14, 18$.

The spectral gap lands exactly on the threshold. For the self-similar variable $D = \varepsilon/3 + (2/3)D'$ the composition operator $\mathcal{C}\varphi(t) = \sum_\varepsilon p_\varepsilon \varphi(\varepsilon/3 + (2/3)t)$ is compact on analytic functions on a disc of radius > 1, since each branch maps it strictly inside itself. Its spectrum is exactly $\{(2/3)^n\} \cup \{0\}$: all three branches share the derivative $2/3$, so $\mathcal{C}(t^n) = (2/3)^n t^n +$ lower order and the operator is triangular in the monomial basis. A Chebyshev discretisation at $24, 40, 60, 80$ nodes returns $1, 0.666667, 0.444444, 0.296296, \ldots$ The second eigenvalue is $2/3$ exactly, so no argument whose only input is this gap can produce the strict inequality.

Phase-blind arguments cannot work. The relevant small-ball quantity has infimum exactly $2/3$ by pigeonhole and Fekete, and attaining that infimum is *equivalent* to $\mu \in L^2$. Every Frostman-type or small-ball argument therefore yields a dimension bound and provably never L^2.

5 Numerical record

All figures come from methods sharing no code: exact prefix densities at depths $20, 24, 26$; a stationary-equation solver on grids of 1.8, 3.5 and 7.1 million cells; a twenty-million-draw Monte Carlo; the closed cosine product; and exact integer counts.

fibre deficits	$j_0 = \log(9/8) = 0.117783$, then 0.002024, 0.000584, $4.95 \cdot 10^{-5}$, $1.35 \cdot 10^{-5}$, $1.30 \cdot 10^{-5}$
their sum	0.120467 nats, equal to $\int f \log f$ to fourteen digits
L^2 norm	$\int f^2 = 1.206310$ and $\sum_{k \geq 1} \lvert\hat{\mu}(k)\rvert^2 = 0.103155$, the two sides meeting as $1 + 2S = \lVert f \rVert_2^2$
octave energy	$E(R) \approx R^{-0.78}$ over octaves 4 to 18, stable across sub-windows
octave maxima	$\max \lvert\hat{\mu}\rvert\, R^{1/2}$ within 0.034–0.073 over octaves 14 to 24, no monotone trend, though the last six drift mildly upward; the truncated P_K grows and is a different object
weight profile	$\mathrm{Cum}(u)(4/3)^{-K} \to \frac{2}{3}\lVert f \rVert_2^2\, u$; measured 0.804446 at $u = 1$
endpoint exponent	$f(x) \approx x^{0.71}$, matching $\log 2 / \log(3/2) - 1 = 0.7095$
window constant	$\int_0^1 \exp(\sum \varphi)$ over $(4/3)^K$ equal to 0.402028 at $K = 16, \ldots, 24$, six digits flat; the remainder bounded
growth constant	$6 \int_0^M Q^2 = 1.205739, 1.206198, 1.206295, 1.206307, 1.206310$ at $M = 10, \ldots, 10^5$, against $1 + 2S = 1.206310$
half-shift, second route	$\int \cos(3\pi v) Q^2 / \int Q^2 = 0.33333349$ against $1/3$, i.e. $\rho(0) = 3\rho(3)$, which is $A(0) = 3A(1/2)$ in ν's coordinates
tail cross-ratio	$0.032, -0.039, -0.008, -0.041$ from $M = 10$ to 10^4; the $1/3$ sits entirely in $u < 10$, which carries 99.95% of the mass
renewal correlation	mean 0.978, s.d. 0.087, s.e. 0.020, slope +0.001 per step over $n = 8, \ldots, 27$; exponent 0.775 against $E(R) \approx R^{-0.78}$
comparability	$C_K = (3/4)^K I_K/(3B_K)$ runs 119.0 to 244.3 over $K = 8, \ldots, 35$, median 187.1, factor 2.05; bounded, not constant
octave rate	per-step rate of I_K over $K = 8, \ldots, 35$: arithmetic 1.0032, geometric 0.9973, OLS 0.9829; sign undetermined, Jensen gap 0.0061
escape rate, $1/\varphi$	exact 0.6202985760 four ways; float filter converges to $0.618014 \pm 0.000039 = \lambda$; smoothing gives λ at every resolution
lines near the target	$\#\{\lvert 3^{n+1} - S_\varepsilon \rvert \leq D 2^{n+1}\} = D\, 2^n$ exactly, $n = 4, \ldots, 16$, matching uniform spreading; nearest line at distance $2^{-(n+1)}$
one-step ratios	0.019, 0.289, 0.085, 0.273, 0.966, 0.007, 0.119, 0.736, 0.533, 0.187

The fifth of those ratios deserves a line: the contraction of the fibre bias is not uniform in n, and any argument assuming it is will fail at level four.